Irreplaceable Middlemen:
Evidence from the Expulsion of Indians from Uganda*

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Abstract

Middleman minorities dominate the trade between producers and markets, yet they are often expelled in the expectation that the majority can easily replace them. I ask whether the majority takes over the economic function of a middleman minority once it is removed, in the context of the 1972 expulsion of Indians from Uganda. Using newly digitized census and road-network data and quasi-random variation from delays in a road-upgrading program, I find that households in localities more exposed to the loss of Indian middlemen obtain less crop value from their farmland four decades later. These losses reflect lower value per cropped acre rather than farming less land, and are not explained by a general contraction of the local economy, a loss of credit and inputs, or out-migration. Consistent with persistent disintermediation, the decline is largest for the most marketed crops, which depend most on intermediation. Moreover, more-exposed households market a smaller share of their harvest and sell less to traders. Lastly, the loss of intermediation has downstream consequences for household welfare: more-exposed households spend less on durable goods. Together, the results suggest that removing a middleman minority destroyed its economic function rather than transferring it to the majority.

Keywords: Middleman minorities, Ethnicity, Expulsion, Development, Persistence
JEL: J15, J24, J44, N47, O12

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1 Introduction

Middleman minorities—ethnic minorities that occupy the commercial layer between producers and markets—are a recurring feature of economies across history (Blalock, 1967; Bonacich, 1973).[1] In settings where formal contract enforcement is weak, middleman minorities screen partners, extend credit, and sanction those who cheat at lower cost than outsiders can (Greif, 1993; Landa, 1981),[2] which gives them a commercial advantage that outsiders could not easily match. While we know much about how middleman minorities can complement the economic activities of the majority and thereby foster economic development (Artiles, 2026; Jha, 2013; Johnson and Koyama, 2017). Much less is known about whether the economic function they perform is substitutable.

This paper asks: when a middleman minority is removed, can others take over the economic function it performed? If middleman minorities persist because barriers keep others out, removing them should open their activities to others and improve the allocation of talent. If instead their position rests on group-specific network capital that cannot be transferred, removing the group may destroy the services it provided. The answer to this question helps inform policies that expel or restrict trading minorities that often presume that their function is replaceable.

In this paper, I take a historical perspective to examine this question. I do so by leveraging the expulsion of ethnic Indian minority from Uganda in 1972.[3] Indians made up less than one percent of Uganda's population, yet they occupied the intermediary layer of the country's agricultural value chain: the aggregation, processing, transport, and wholesale trade that connected African smallholders to distant markets. In 1959, more than half of economically active Indians worked in commerce (East African Statistical Department, 1960), while Africans were mostly smallholder farmers (De Haas, 2022). Although African traders outnumbered Indian traders by more than two to one in the early 1950s, they handled about a third as much business (Uganda Government Printer, 1955).

[1] Prominent examples of middleman minorities include Jewish moneylenders and merchants in medieval Europe (Becker and Pascali, 2019), Armenian and Greek traders in Anatolia (Arbatlı and Gokmen, 2023), Chinese merchants in Southeast Asia (Do et al., 2025), Lebanese traders in West Africa, and Indian traders in East Africa.

[2] Middleman minorities owe their position partly to the skills migrants bring from their places of origin and partly to ethnic networks. In some cases, an orientation toward eventual return further strengthens in-group solidarity and keeps the group apart from the host society (Bonacich, 1973).

[3] In colonial and early post-independence Uganda, this population was also commonly referred to as "Asian," the term used in official statistics and public debate. Throughout the paper, I use the term "Indian" to refer to people who migrated to East Africa from the Indian subcontinent and their descendants, whose origins lie in present-day India, Pakistan, and Bangladesh.

Indian intermediation ran on ethnic networks. Merchants recruited newcomers from their home districts in India, staffed their up-country stores with kin and co-religionists, and set new arrivals up in business with goods advanced on credit. The *dukawallah*, or shopkeeper, drew his stock on credit from a co-ethnic wholesaler and advanced crop finance to the farmers he bought from. The wholesaler extended that credit not on collateral but on the shopkeeper's standing within the community. The same exclusivity that made informal contracts enforceable also raised the barriers to entry facing Africans. In August 1972, President Idi Amin ordered Uganda's Indian population of approximately 74,000 to leave within ninety days, justifying the measure as an "economic war" to return commerce to African hands (Jamal, 1976).

This paper exploits granular spatial variation in the sudden loss of middlemen in the agricultural value chain induced by the Indian expulsion in 1972 to investigate its long-term impacts on African farmers. The outcome of interest is the crop value of farmland which captures the value farmers obtain from their land. This is measured by summing each household's harvest of every crop and valuing it at the crop's national median sales price. I then explore mechanisms behind these effects and downstream impacts on household consumption.

I construct a measure of exposure to the loss of middlemen, building on the fact that Indian settlement was largely concentrated in urban towns.[4] I digitize the 1959 non-African census, which records the Indian population of each of Uganda's 94 towns, and geocode the towns. I also collect data on the 1961–1974 road network, recording each segment's surface class in every year. Combining these two sources, I measure each locality's exposure to the expulsion as its Indian market access in 1971, the year before the expulsion. In the spirit of Donaldson and Hornbeck (2016), it is computed as the sum of the Indian populations of all towns, each discounted by the shortest-path travel cost between the locality and the town on the 1971 road network.

I compute this measure for two sets of locations, which correspond to the two levels of my analysis. The first consists of survey localities, the communities sampled in the 2011/12 Uganda National Panel Survey (UNPS), a nationally representative household survey that records crop output, input use, sales, and the type of buyer for 1,811 farming households in 417 localities. The second consists of historical settlements, the 3,429 populated places listed in the 1964 gazetteer of Uganda, which I digitize and geocode.

[4]Colonial land law restricted non-African access to African land. Trading licenses were granted only in gazetted townships and trading centers, so Indian households and businesses clustered in towns from which they served the surrounding countryside. About 74 percent of Indians resided in towns in 1959 (East African Statistical Department, 1960).

I link these settlements to satellite measures of vegetation from 1981 to 2008 and trace the evolution of agricultural activity over the decades that followed. Throughout the analysis, I exclude households in localities that lie within these towns to remove the direct effect of the departure of the Indian population itself.

A naive comparison of more- and less-exposed localities would be biased, because localities well connected to Indian-populated towns differ systematically from more remote ones in ways that independently affect long-term agricultural outcomes. For identification, I exploit the timing of a road upgrading program, following the recentering approach of Borusyak and Hull (2023). Road upgrades lower the travel cost to Indian-populated towns and exposure to the loss of Indian middlemen after 1972. Starting in 1963, 291 road segments were upgraded under a program designed by a World Bank mission, which selected road segments with traffic above a given threshold and therefore favored roads serving major commercial centers (World Bank, 1962). However, which of these segments were completed before the expulsion is plausibly exogenous. The schedule was revised continuously, and engineering constraints delayed upgrades beyond 1972.

Following Borusyak and Hull (2023), I view the realized completion of upgrades by 1971 as one draw from this assignment process. I simulate 1,000 counterfactual draws by reshuffling which segments were completed by 1971 within strata defined by road class and centrality in the 1961 network, holding the number of completed segments fixed, and compute each locality's expected exposure across these draws. Controlling for expected exposure absorbs the variation that arises from a locality's position in the road network, so that the effect of exposure is identified only from which planned upgrades happened to be completed before the expulsion.

For identification checks, I show that geographic characteristics such as topography and climate are balanced across localities with different levels of recentered exposures. Crucially, important pre-existing economic characteristics such as the density of populated settlements, distance to Kampala, and market access in 1961 are balanced. Moreover, I show that recentered exposure does not predict shocks that occurred after the expulsion and could independently affect long-term agricultural outcomes. This supports the assumption of the exclusion restriction. The timing of road upgrades affects outcomes only through exposure to the loss of middlemen rather than through other changes that followed it.

I use a simple conceptual framework to guide my empirical investigations. Farmers allocate land across crops according to the wedge between the price a crop commands

in the market and what they receive at the farm gate. This wedge reflects the cost of the intermediation services that carry crops to buyers. Before 1972, these services were provided by Indian traders and, to a lesser extent, by African traders. Localities differed in how much they relied on each of them. The expulsion has an ambiguous effect on this wedge. On the one hand, localities that had depended more on the Indian commercial network may have been left with fewer providers of intermediation services once the Indians departed. On the other hand, the same localities may have found it easier to rebuild intermediation. If dependence dominates, more-exposed localities face a wider wedge today and should have a lower crop value of farmland, concentrated among the most marketed crops, with sales shifting away from traders. If replacement dominates, these predictions are reversed.

Four decades after the expulsion, households in localities more exposed to the loss of Indian middlemen obtain significantly less crop value from their farmland. A one-standard-deviation increase in recentered exposure lowers crop value of farmland by about 17 percent. This corresponds to a locality whose road to the nearest Indian-populated town was upgraded before the expulsion, reducing travel costs by about 16 kilometer-equivalents more than its position in the road network would predict. The effects are economically large: for an average farming household, this amounts to about 289,000 Uganda shillings per year.

These estimates are robust to a range of checks: alternative definitions of the counterfactual road networks, restricting the comparison to localities with similar geography, alternative measures of exposure (including exposure to the nearest Indian-populated town and decay parameters from 10 to 50 kilometers), alternative approaches to inference including different distance thresholds, and dropping each region or sub-region one at a time. The effect is specific to exposure to Indian commerce rather than to towns in general. Among households whose nearest town had few Indians, the timing of road upgrades has no effect on crop value of farmland. Finally, satellite data allow me to trace agricultural activity over time. More-exposed settlements have lower vegetation throughout the period from 1981 to 2008, suggesting a lasting reduction in agricultural production that was already visible by 1981.

I rule out four alternative explanations: (i) differences in land use, (ii) a general contraction of local economy, (iii) a loss of credit and inputs, (iv) selective out-migration, and (v) loss of human capital.

First, a lower crop value of farmland may reflect that more-exposed households farm smaller plots or leave more of their land fallow, for reasons unrelated to exposure to

Indian middlemen. I next decompose the effect into the size of farmland, the share of farmland that is cropped, and the crop value per cropped acre. I find that the decline in crop value of farmland is not explained by farming less land or by cropping it less intensively.

Second, the departure of Indian households may have depressed economic activity around their towns despite excluding localities that fall within towns throughout the analysis. I show that the estimates remain similar when I exclude localities at varying distances from these towns. Furthermore, there is no detectable effect on nighttime lights, suggesting no broad impact on non-farm activity.

Third, Indian traders may have supplied credit and inputs to farmers. I find no effect on farm equipment or on the use of purchased inputs. Fourth, recentered exposure does not predict out-migration. The results hold among household heads who have lived in their locality since before the expulsion.

Lastly, the expulsion may have disrupted schooling, so that farmers today are less productive because of lost human capital. Using a Duflo-style cohort difference-in-differences, I find that cohorts of primary-school age or younger in 1972 completed less schooling in more-exposed districts of birth, but this did not translate into differences in employment sector, suggesting that the loss of human capital is unlikely to explain the main results.

I hypothesize that the decline reflects persistent disintermediation: the persistent loss of the intermediation services that had connected farmers to markets. These intermediation services might have been difficult to replace because they rested on group-specific network capital. Linking farmers to distant markets requires a chain of buyers, transporters, processors, and wholesalers who extend credit to one another and trust each other. I test two implications of this hypothesis.

First, if the expulsion raised the cost of reaching distant markets, its effect should be concentrated among the crops that rely most on intermediation services. To test this, I rank crops by their total sales value at national median prices, as a proxy for their reliance on intermediation, and estimate the effect of exposure on the value of nested baskets of the most marketed crops. The three most marketed crops depend on the intermediation: coffee must be hulled and graded before it is sold to exporters, maize must be aggregated from many small producers and stored, and food bananas are bulky and perishable, so they need to be collected and transported to towns. I show that an increase in exposure to the loss of Indian middlemen significantly lowers the value of every basket, and that the effect is largest for the three most marketed crops: coffee,

maize, and food bananas.

Second, if fewer intermediaries operate, less of what households produce should reach the market. I show that households in more-exposed localities are modestly less likely to sell any crop and they sell a smaller share. Moreover, households in more-exposed localities are less likely to sell to traders and more likely to sell to consumers, neighbors, and other buyers, and the share of their sales going to traders is lower. These results indicate that in more-exposed localities, farmers produce less of the crops whose value depends on reaching buyers, sell less of what they produce, and sell more of it through local channels that bypass traders.

Finally, I provide suggestive evidence that persistent disintermediation has downstream consequences for household consumption. Lower crop value translates into lower household spending on durable goods such as education, clothing and footwear, and insurance. In contrast, I find no significant effect on food consumption or on total consumption. This pattern is consistent with households preserving their food consumption through home production, since they sell a smaller share of their harvest, while absorbing the loss in crop value by cutting back on durable purchases.

Related literature. This paper contributes broadly to the literature on the economic consequences of expelling high-skilled minorities. Much of this literature concerns Jewish minorities in Europe and Russia (Arbatlı and Gokmen, 2023; Becker et al., 2024; Huber et al., 2021; Waldinger, 2010, 2012). A central theme of this work is that the expellees held formal, transferable skills, as scientists, professors, or firm managers, so that their expulsion affected economic outcomes through the loss of human capital. Relative to this literature, this paper differs in that the expulsion removed intermediation, which farmers in agricultural economies rely on to reach markets, with long-term effects on agricultural outcomes. A few closely related studies focus explicitly on middleman minorities in trade. Grosfeld et al. (2013) show that in the former Pale of Settlement, where Jews had specialized in commercial occupations before the Holocaust, residents today hold more antimarket attitudes and are less entrepreneurial. Instead of focusing on cultural outcomes, I focus on economic outcomes. Acemoglu et al. (2011) exploit variation in the intensity of the Holocaust across Russian regions, which depended on both the pre-war Jewish population and German occupation. Their estimates therefore capture the combined effect of occupation and the loss of Jewish middlemen on aggregate economic and political outcomes. I instead isolate the loss of middlemen, in an expulsion without a concurrent war or occupation, and examine, at a micro level, the effects on the producers who relied on them.

Second, this paper relates to the literature on the economic impact of ethnic diversity (Alesina et al., 2003; Easterly and Levine, 1997; Montalvo and Reynal-Querol, 2005). One view holds that groups competing for the same resources and occupations are prone to conflict. Another holds that groups specializing in different activities can gain from exchange, and that such complementarity can sustain both economic prosperity and tolerance (Artiles, 2026; Becker and Pascali, 2019; Do et al., 2025; Jha, 2013, 2018; Kok et al., 2026), although it can break down when political shocks interrupt the relationships between groups (Grosfeld et al., 2020). The findings of this paper suggest that removing a complementary group can destroy the gains from exchange rather than transfer them to the majority.

The remainder of the paper is organized as follows. Section 2 provides historical background. Section 3 describes the data. Section 4 describes the identification strategy. Section 5 presents the main results. Section 6 presents evidence on the mechanisms. Section 7 examines downstream impact. Finally, section 8 discusses and concludes.

2 Historical Background

2.1 Indians in Uganda

This section describes the population expelled in 1972, that is, the ethnic Indian minority. On the eve of the expulsion, Uganda's Indian population numbered 74,308 according to the 1969 census, less than one percent of the country's population, of whom 26,657 held Ugandan citizenship.[5]

Indian merchants had traded on the East African coast for centuries before colonial rule, but their large-scale settlement in the interior came in two major waves. The first consisted of indentured laborers recruited from British India to build the Uganda Railway, which linked the port of Mombasa to Lake Victoria between 1896 and 1901 (Mangat, 1969). Of the 31,983 Indian laborers brought to East Africa for its construction, 2,493 died and 6,454 were invalided home, while most of the remainder returned to India and 6,724 chose to stay in East Africa (Mangat, 1969).

From the early twentieth century, and especially between the 1920s and the 1950s,

[5]In the census 1969, share of ethnic Indians (the sum of census "Indians", "Pakistanis", "Goans" categories) constituted 0.78% of the 9,535,051 total population in Uganda (Uganda. Ministry of Planning and Economic Development. Statistics Division, 1971). Estimates for the number of Indians in 1972 vary widely. Hundle (2013) estimates 55,000 at the time of the decree, while other accounts give about 80,000 (Jørgensen, 1981).

free, or "passenger," migrants paid their own passage to East Africa, drawn largely from Gujarat, Kutch, and Kathiawar, together with smaller numbers of Punjabis and Goans (Mangat, 1969; Morris, 1956). Many belonged to communities with long-established commercial networks, including Hindu Bhatias and Lohanas and Muslim Ismailis, Bohras, and Ithna'asheris (Oonk, 2013).[6] The Indian communities remained endogamous (Morris, 1956). Appendix Figure A1 shows that the Indian population grew rapidly from about 5,000 in 1921 to 35,215 in 1948 and to 71,933 by 1959. It then stabilized, reaching 74,308 in 1969.

Despite making up less than one percent of Uganda's population, the Indians came to dominate its commerce. According to the 1959 census of the non-African population, only about 87,000 of Uganda's residents were non-Africans, of which Indians were the largest group among them, accounting for 82.6 percent, followed by Europeans (12.5 percent), Arabs (2.2 percent), people of mixed descent (1.5 percent), Somalis (0.5 percent), and others (0.6 percent) (East African Statistical Department, 1960). Among economically active and employed Indians, 53.3 percent worked in commerce, followed by manufacturing (10.9 percent) and agriculture and mining (10.4 percent), with smaller numbers in public service and the professions. The European workforce, by contrast, was concentrated in public service (39.9 percent), the professions (15.3 percent), and religious organizations (14.5 percent). Africans were overwhelmingly smallholder farmers, who grew the cotton and coffee on which the colonial economy depended (De Haas, 2022).[7]

Indians' position in Uganda's agricultural economy was concentrated in the segments of the value chain that connected African farmers to distant markets, that is, buying and aggregating crops, processing, transporting, and trading them wholesale.

These activities rested on the social organization of the Indian communities. First, Indian firms recruited and trained their staff through kinship. Indian traders who had settled in Uganda brought their relatives from India to help expand their businesses

[6]The Bhatias and Lohanas are Hindu mercantile castes largely from Kutch, Kathiawar and Sindh, with long histories in Indian Ocean trade. The Ismaili and Ithna'asheri Khojas descend from Hindu trading groups that converted to Islam.

[7]Indians were considerably more successful than their African counterparts. Indian households earned on average fourteen times African household income (De Haas, 2022). In 1952, while licensed African traders outnumbered Indian traders by more than two to one ratio, they collectively handled only about a third as much business. The Land Transfer Ordinance of 1906 prohibited the transfer of African land to non-Africans without the Governor's consent, which prevented Indians from becoming smallholder or estate farmers, with the exceptions of Crown leases and freehold estates. Moreover, the rules governing crop marketing kept Africans out of processing. Cotton could only be bought at gazetted markets by licensed buyers. Since ginnery licenses were held in practice only by Europeans and Indians, Africans were not able to gin cotton until the reforms of the early 1950s. Together, these forces channelled Indians into the part of the agricultural economy that connected African farmers to markets.

(Oonk, 2013). New arrivals often served an apprenticeship in a relative's shop before setting up on their own. Second, goods and credit flowed along chains of traders linked by these ties. For instance, import and wholesale houses in Mombasa and Kampala supplied up-country shops on credit and shopkeepers in turn advanced goods to farmers against the coming harvest and buying their crops in return. The kinship ties give the Indians an advantage in intermediation.

In settings where contract law is weak, an ethnically homogeneous middleman group is particularly effective at economizing on the costs of enforcing contracts (Landa, 1981). Shared kinship and ethnic codes act as a club, within which members can screen trading partners and sanctions those who cheat. Since middleman minorities are likely oriented toward eventual return rather than permanent settlement, this keeps them apart from the host society and strengthens their in-group identity and solidarity (Bonacich, 1973). These mechanisms are valuable in intermediation where transactions pass through several agents in different locations, payment is often deferred until goods have been resold, and the quality of goods is difficult to verify at each stage.

2.2 The 1972 Expulsion

On 4 August 1972, President Idi Amin announced that Uganda's Indians had ninety days to leave the country. Although the order initially targeted non-citizens, it was applied in practice to the entire community, including those who held Ugandan citizenship. By 1973, no more than about 1,000 remained in Uganda (Oonk, 2013). President Amin presented the expulsion as the opening of an "economic war", whose purpose was to transfer the country's commerce from Indians into the hands of Africans. How the expulsion impacted agricultural economy depends on the role of Indians in the agricultural value chain. I describe below the services that this chain required and which of them remained in Indian hands prior to the expulsion.

A marketed crop needs to pass through three stages of value chain. First, upstream services that reach farmers before and during production, namely, credit, seeds, and tools, and the consumer goods that farmers buy on credit against the coming harvest. Intermediation services connect the farm to distant markets, including aggregation, processing, transport, and distribution. Finally, the downstream services are exports.

Before the 1950s, Indian firms operated at most every stage of this chain. The expulsion was, however, the culmination of a longer process of Africanization policies. For more than two decades before 1972, colonial and post-independence governments pur-

sued Africanization polices that transferred the two ends of the chain from Indians to the state and to state-supported cooperatives. For instance, the Lint Marketing Board took over cotton exports from 1949 and the Coffee Marketing Board obtained a monopoly over coffee exports in 1969. Cooperative unions, supported by government loans and the purchase of Indian-owned ginneries after 1952, also took over cotton ginning, and by 1970 they handled essentially the entire cotton crop (Young et al., 1981). In terms of upstream services, the state created alternative sources of agricultural credit, beginning with the Uganda Credit and Savings Bank in 1950 and later through cooperative credit societies.

What remained in Indian hands on the eve of the expulsion was largely the intermediation—the services that transport crops and goods between rural areas and towns. Indian firms still dominated urban wholesale trade and import distribution, supplied African retailers on credit, ran road haulage, and owned many agro-processing and manufacturing firms. The marketing of food crops such as maize, beans, and groundnuts, continued to rely on private buyers and wholesalers.

2.3 Conceptual Framework

A farmer i in locality ℓ allocates each plot of land to one of several crops. A marketed crop k sells at price p_k, which is common across localities because marketed crops are traded nationally or exported. To sell it, the farmer pays a wedge, $\tau_{k\ell} = \theta_k \left(c_\ell + m_\ell\right)$, where c_ℓ is the cost of the intermediation services that carry the crop to its buyer (aggregation, processing, storage, transport, and wholesale); m_ℓ is the margin that the providers of these services keep. The parameter $\theta_k \in [0, 1]$ captures how much crop k relies on the intermediation. A plot is devoted to crop k if its net return $p_k - \tau_{k\ell}$ exceeds the value v_i of its best alternative use, which varies across plots with land quality.

Let E_ℓ denote a locality's pre-expulsion exposure to the Indian commercial network. The expulsion removed Indians from every town, but the impact differed across localities. To capture this, let S_ℓ denote the intermediation capacity available to farmers in locality ℓ after 1972. A greater intermediation capacity reduces both c_ℓ and m_ℓ.

The post-expulsion intermediation capacity therefore depends on exposure as follows: $S_\ell = S_0 - \delta E_\ell + \rho E_\ell$, where S_0 is the intermediation capacity that a locality would have had despite exposure to the Indian network (e.g., provided by African traders and cooperatives). δE_ℓ captures *dependence channel*, where fewer African providers fill the gap left by the departed Indians; ρE_ℓ captures *replacement channel*, where new

African providers rebuild the intermediation after the expulsion. The expulsion may have left behind resources that new providers use to rebuild the intermediation such as physical assets. Moreover, Africans who had worked for or alongside Indians may have learned commercial practices and skills.

The net effect of exposure on intermediation capacity is $\frac{\partial S_\ell}{\partial E_\ell} = \rho - \delta$. The effect of exposure on wedge for crop k is $\frac{\partial \tau_{k\ell}}{\partial E_\ell} = \theta_k \underbrace{\left(\frac{\partial c}{\partial S} + \frac{\partial m}{\partial S}\right)}_{<0}(\rho - \delta)$. If dependence channel dominates ($\delta > \rho$), more exposed localities were left with less capacity after 1972 and face a wider wedge. If replacement channel dominates ($\rho > \delta$), they face a narrower one.

I use this framework to guide my empirical investigation. First, I conceptualize the 1972 expulsion as a shock to the intermediation capacity described above. By removing Indians from every towns, it changed the capacity available to farmers after 1972 in proportion to each locality's pre-expulsion exposure to the Indian commercial network. Second, I map the framework's core concepts to empirical measures. Third, my empirical analysis focuses on the following set of testable predictions:

Prediction 1 (Crop value of farmland): If dependence dominates, greater pre-expulsion exposure lowers the crop value of farmland. If replacement dominates, it raises the crop value of farmland.

Prediction 2 (Crop composition): Since the effect scales with θ_k, effects are stronger among the most marketed crops that rely more on the intermediation and weaker for crops that rely less on the intermediation.

Prediction 3 (Market participation): If dependence dominates, greater pre-expulsion exposure may reduce participation in crop markets, as farmers shift land toward crops consumed at home. If replacement dominates, the effect is opposite.

Prediction 4 (Composition of buyers): If dependence dominates, greater pre-expulsion exposure shifts sales away from traders toward other buyers, such as consumers, neighbors, and relatives. If replacement dominates, the effect is opposite.

3 Data

The analysis relies on data constructed at two levels. The first is a household-level dataset on contemporary agricultural outcomes, drawn from the 2011/2012 wave of the Uganda National Panel Survey (UNPS), in which households are nested within survey localities. The second is a settlement-level dataset, which I digitized and geocoded from the 1964

gazetteer (Office of Geography, 1964). Below, I first describe the treatment variable and then each dataset.

3.1 Indian Market Access in 1971

I measure each locality's exposure to the loss of Indian middlemen as its market access to Indian-populated towns, in the spirit of Donaldson and Hornbeck (2016). Their measure sums the populations of all other locations, each discounted by the cost of reaching it. I adapt this measure in two ways: I weight towns by their Indian population rather than their total population and discount them exponentially in travel costs.

The town data come from the 1959 census of Uganda's non-African population (East African Statistical Department, 1960), which records the Indian population of each of Uganda's 94 towns. I digitize these records and geocode each town. About 74 percent of Indians lived in these towns in 1959. The road data come from Jedwab and Storeygard (2022), which record the location of each road segment and its surface class (paved, improved, or earth) in every year.

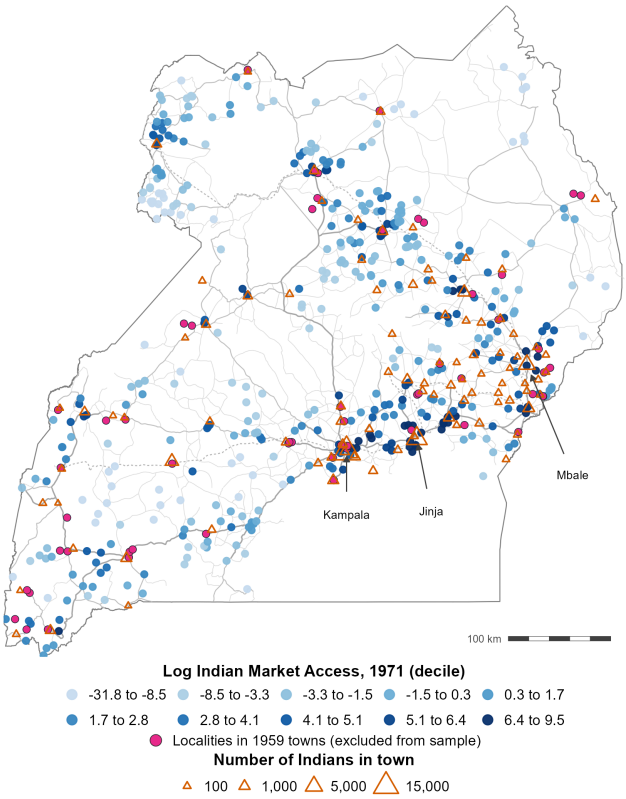
Indian market access for locality l in 1971 is

$$IndianMA_{l,1971} = \sum_{j=1}^{N} \frac{IndPop_j}{\tau(Network_{1971}, Loc_l, Loc_j)} \tag{1}$$

where N is the total number of Indian-populated towns in 1959, $IndPop_j$ is the size of Indian population in town j in 1959, $\tau(Network_{1971}, Loc_l, Loc_j)$ is a function of the travel cost between l and j based on the state of road network in 1971, which is $\exp(0.1\ \tau_{lj,1971})$.[8] $\tau_{lj,1971}$ is the travel cost using the shortest path between a locality and an Indian-populated town using Dijkstra's algorithm. I construct this measure for the two sets of locations in which I observe outcomes: the survey localities of the 2011/12 UNPS, used in the household-level analysis, and the historical settlements of the 1964 gazetteer, used in the settlement-level analysis. Figure 1 illustrates the variation across survey localities. I now describe these two datasets in turn.

[8]The specific functional form follows Borusyak and Hull (2023). The value 0.1 corresponds to a distance decay parameter of 10 kilometers. This implies that the weight an Indian-populated town receives falls by a factor of e for every additional 10 kilometer-equivalents of travel. I choose a steep decay because intermediation between Indian traders and African farmers was local. In robustness checks, I vary the distance decay parameters to 20km, 30km, 40km, and 50km. Results are not sensitive to the choice of distance decay parameters.

Figure 1: Indian Market Access in 1971 at the Locality-Level



Notes: The figure maps log-transformed Indian market access in 1971, plotting the UNPS localities. Points are shaded by decile; grey lines show the 1971 road network. Magenta points lie inside a 1959 town and are excluded from the estimation sample. Orange triangles mark the 1959 towns, sized by their Indian population.

3.2 Household-level Data

My household data come from the 2011/12 wave of the Uganda National Panel Survey (UNPS), conducted by the Uganda Bureau of Statistics as part of the World Bank's LSMS-ISA program. The UNPS is nationally representative, and its agricultural module records, for each crop and cropping season, how much the household harvested and the land it cultivated, how much it sold, and to whom, as well as the use of chemical inputs. The richness of data allows me to study not only the crop value of farmland but also the channels through which exposure affects it. The estimation sample consists of 1,811 farming households in 417 localities. Appendix Table A3 reports summary statistics.

Farmland and crop value. I measure the size of farmland as the total area of parcels the household cultivates or leaves fallow, using GPS measurements where available and farmer reports otherwise. To measure the crop value of farmland, I sum harvests over

all crops and both seasons of 2011, valuing each crop at its national median sale price.

Cropped area and crop value per cropped acre. To decompose the crop value of farmland, I measure cropped area as the area of farmland planted, averaged over the two cropping seasons. Crop value per cropped acre is the crop value of farmland divided by cropped area.

Crop value and harvest by crop. To examine which crops drive the decline in crop value, I construct the crop value of nested brackets of the most marketed crops: the top 3, top 5, and top 10 crops by total sales value. For each basket, I sum the value of the harvest of the crops in the basket, valued at their national median prices. I also measure the quantity harvested of each of the main crops, in kilograms, summed over both seasons.[9]

Market participation. I measure participation in crop markets with two variables: an indicator for whether the household sold any crop in either season and the percentage of the value of its harvest that it sold.

Composition of buyers. To measure whom households sell their crops to, I use the survey question on who bought the largest part of each crop sold. I group the buyers into two types: traders, who are private traders in a local village or district market and non-traders who are consumers at market, neighbours and relatives, government and other buyers. Since the survey names only the main buyer, I assign the whole sale of a crop to that buyer's type. For each household that sold at least one crop, I construct two indicators, one for selling any output to a trader and one for selling any output to a non-trader. I also sum the value of sales to each type across crops and seasons, which gives the value of crops sold to traders and the value sold to non-traders.

Household consumption. I use the consumption aggregates constructed by the Uganda Bureau of Statistics, expressed as monthly consumption per adult equivalent and deflated to 2005/06 prices.

Household controls. I use three characteristics of the household from the roster and use them as household-level controls. They are whether the head is female, the head's age, and the number of members in the household.

[9] Appendix Table A5 presents the ranking of crops. The top five crops are coffee, maize, food bananas, sugarcane, and beans, which account for about 60 percent of total sales value.

3.3 Settlement-level Data

I use the 1964 gazetteer of Ugandan populated places (Office of Geography, 1964), which I digitize and geocode to obtain 3,429 historical populated settlements. Since these settlements are observed prior to the expulsion, this helps mitigate endogeneity concerns.

Agricultural activity. I measure agricultural activity using the normalized difference vegetation index (NDVI), which compares reflected near-infrared and visible red light and increases with the density of living vegetation. NDVI has been used as a proxy for agricultural production where direct measures are unavailable (Asher and Novosad, 2020). For each settlement, I take the annual mean index in the satellite cell containing each settlement and assign it to the settlement. I average those values within seven four-year periods spanning 1981 to 2008 to reduce measurement error from year-to-year fluctuations in weather. Since NDVI measures the density of vegetation rather than the quantity or value of crops, I interpret these estimates as evidence on the direction and persistence of the effect on agricultural activity.

Non-agricultural activity. I use nighttime lights from the Defense Meteorological Satellite Program's Operational Linescan System (DMSP-OLS) for 1992, 2000, and 2010. For each settlement, I construct an indicator for whether the mean digital number within 5 kilometers of the settlement is positive. As an alternative measure, I compute the share of lit cells within 5 kilometers of the settlement.

3.4 Other Controls

I include two sets of controls, which capture differences in agricultural potential and in pre-expulsion economic development that may have shaped Indian settlement patterns and agricultural outcomes today. I measure controls for both survey localities and historical settlements.

Agroclimatic controls. I control for terrain ruggedness using data from Nunn and Puga (2012) since steep terrain likely raises the cost of cultivating land and of moving goods off it. Second, I control for cotton and coffee suitability, using data from FAO Global Agro-Ecological Zones (FAO and IIASA, 2021), since cotton was the dominant crop and may predict both agricultural potential and Indian settlement. Third, annual rainfall from 1960–1969 comes from Fick and Hijmans (2017), which is constructed by first summing monthly totals and then averaging over the ten pre-expulsion years.

Geographic controls. I control for the number of populated settlements within 10

kilometers of a locality from Office of Geography (1964) to account for pre-expulsion population density. Second, I control for the distance to the nearest Indian-populated town in 1959, which captures proximity to urban markets. Third, I control for the distance to Kampala, the capital. Fourth, I control for the distance to the nearest Christian mission station, using data from Nunn (2010).

4 Empirical Strategy

A naive comparison of more- and less-exposed localities is biased because localities better connected to Indian-populated towns differ systematically from more remote ones in ways that independently affect long-term agricultural outcomes. To isolate variation in exposure that is unrelated to these advantages, I exploit the timing of a road upgrading program, following the recentering approach of Borusyak and Hull (2023).

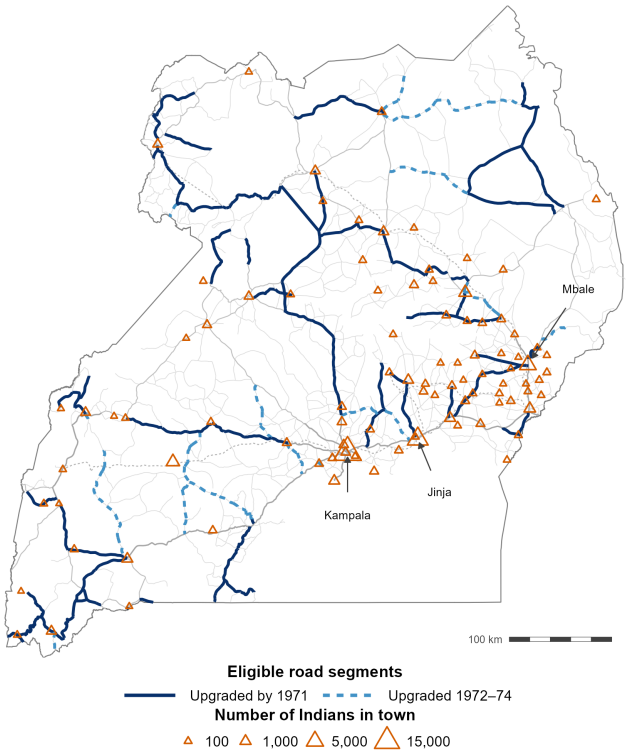
Between 1963 and 1974, 291 road segments were upgraded under a program designed by a World Bank mission, which selected road segments carrying traffic above a given threshold and therefore favored roads serving major commercial centers (World Bank, 1962). 222 were completed by the end of 1971 and the remaining 69 between 1972 and 1974 (see Figure 2).

Selection of road segments into the program is not random but which selected segments were completed before the expulsion is plausibly random. Contemporary assessments suggest that the sequencing within the programme was not itself planned. The government committed to a number of miles to be upgraded within a fixed expenditure envelope, but which segments filled that target remained under continuous review, and secondary roads were programmed piecemeal by local administrations with little technical supervision (World Bank, 1962). Completion dates were further governed by engineering constraints.[10]

Figure 3 shows standardized mean differences in pre-treatment characteristics between the eligible road segments upgraded by 1971 and those upgraded between 1972 and 1974. The mean differences are statistically indistinguishable from zero in the characteristics of road segments such as length, sinuosity, average slope and terrain ruggedness, but not in their position in the road network and in the road class. Road segments finished before 1972 were more central in the 1961 road network. I therefore treat com-

[10]For instance, bituminizing a road was uneconomical unless it already lay on its final alignment, so upgrading frequently required realignment that included survey, land acquisition, and new construction, each of which could postpone completion.

Figure 2: Road Segments Upgraded By 1971 Versus Upgraded Between 1972 and 1974

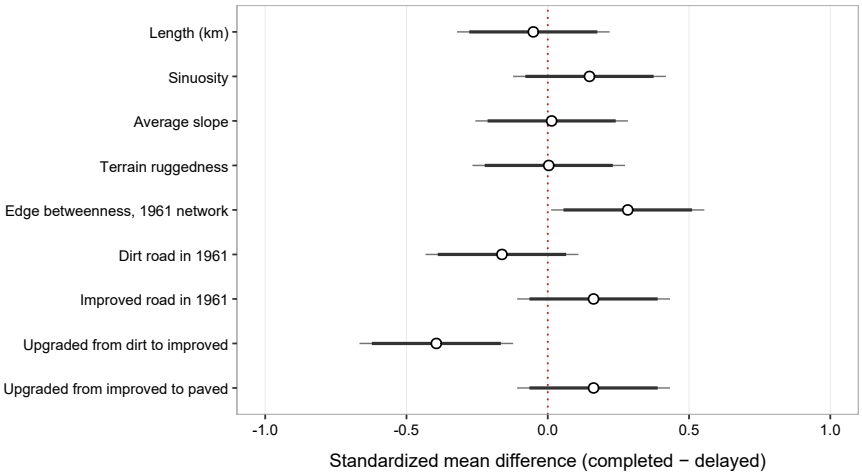


Notes: The figure illustrates the 1971 road network of Uganda. Eligible road segments are the 291 segments whose road class improved between 1961 and 1974. 222 were upgraded by 1971, shaded (solid blue lines) and 69 were upgraded between 1972 and 1974 (dashed blue lines). Triangles mark the 93 towns with an Indian population in the 1959 census, sized by their number of Indians.

pletion as good as random only among segments that are similar on these two characteristics. I define strata by the interaction of segment's road class and the tercile of its 1961 edge betweenness.

Given this assignment process, I construct counterfactual road networks by randomly reassigning which segments were completed by 1971 within each stratum, holding the number of completed segments in each stratum fixed. I draw 1,000 such counterfactual networks, recompute Indian market access in 1971 for every locality under each draw, and define expected exposure as the average of Indian market access across the 1,000 draws. Expected exposure captures the part of a locality's exposure that is predictable from its position in the road network and the set of segments eligible for upgrading. Controlling for expected exposure removes this predictable component, so that the re-

Figure 3: Standardized Mean Difference Between Upgraded By 1971 and Delayed Roads



Notes: Standardized mean difference in pre-existing characteristics between road segments that saw upgrade completed by 1971 versus those that witnessed delays.

maining variation in exposure comes only from which segments happened to be completed before the expulsion. Figure 4 shows the variation of realized, expected, and recentered Indian market access in respective panels.

4.1 Estimating Equation

Throughout the analysis, I impose two sample restrictions. First, I restrict the sample to farming households since the outcomes are defined only for households engaged in crop production. Second, I exclude localities that lie within an Indian-populated town in 1959 to remove the direct effect of the departure of the Indian population itself.[11]

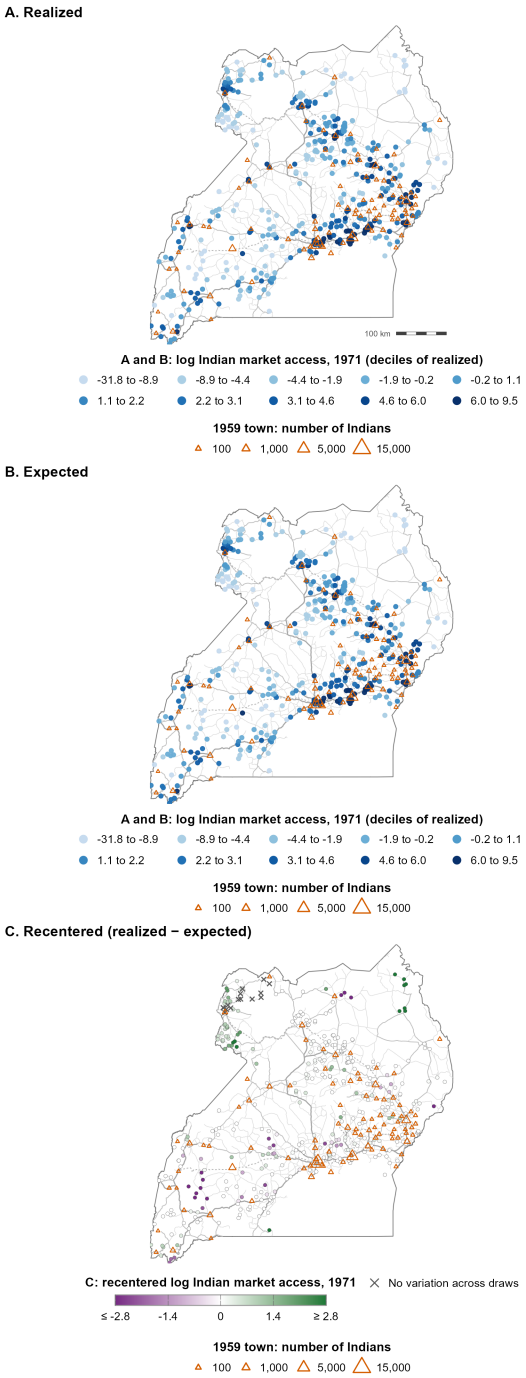
The estimating equation is:

$$Y_{il} = \beta_1 \text{Realized Indian MA}_{l,1971} + \beta_2 \text{Expected Indian MA}_{l,1971} + \mathbf{X}_l'\gamma + \mathbf{X}_i'\zeta + \epsilon_{il} \quad (2)$$

where Y_{il} denotes the outcome of a household i in locality l (e.g., log crop value of farmland), the vector $\mathbf{X}_l$ includes an array of potential confounders that may have affected Indian sorting and downstream development at the locality-level. These include (i) historical economic factors such as settlement density in 1964, distance to the nearest towns in 1959, distance to Kampala, and distance to the nearest mission station; and

[11] In Appendix Figure A7, rows 10 through 13, I show that results are not sensitive to the further exclusion of localities within 5km, 10km, 20km, and 30km of these towns.

Figure 4: Indian Market Access in 1971: Realized, Expected, and Recentered



Notes: The figure shows realized Indian market access in 1971 in Panel A, expected Indian market access in 1971 in Panel B, and the recentered Indian market access in 1971 in Panel C across the UNPS localities. Triangles mark the 93 towns with an Indian population in the 1959 census, sized by their number of Indians.

(ii) agroclimatic factors that capture differences in agricultural potential such as ruggedness, high-input rainfed cotton and coffee suitability, mean rainfall over 1960–1969, the

vector $\mathbf{X}_i$ accounts for household-level differences such as household head's gender and age, household size. Standard errors are clustered based on Conley (1999) that allows for spatial correlation among observations within 50 kilometers.[12] In addition, I also report valid confidence intervals based on the randomization test statistic proposed by Borusyak and Hull (2023).

The parameter of interest, β_1, measures the effect of a one-log-point increase in Indian market access in 1971 on the outcome, holding expected exposure and the controls fixed. Because expected exposure absorbs the component of exposure that is predictable from the road network, β_1 is identified only from variation in which segments happened to be completed before 1972.

Balance checks. Within strata of similar road segments, which eligible road segments were completed by 1971 is as good as randomly assigned. I check whether recentered exposure is balanced on pre-existing characteristics.

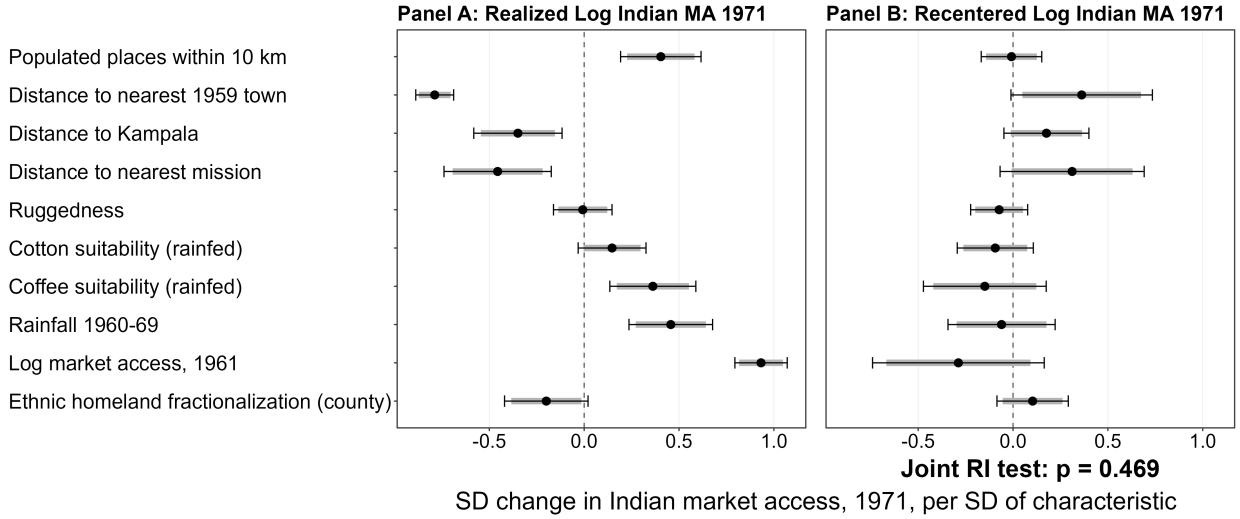
Panel A of Figure 5 first quantifies the identification concerns by regressing realized Indian market access on pre-determined characteristics. Indeed, localities with greater realized Indian market access in 1971 are systematically more populated in 1964, closer to a town in 1959, closer to Kampala, and closer to the nearest mission station, more suitable for coffee cultivation, receive more rainfall, and have greater market access in 1961. Panel B of Figure 5 shows that the recentered exposure, the difference between realized Indian market access and expected Indian market access, is not predicted by any of the pre-determined characteristic.

Exclusion restriction. The road upgrade timing should only affect outcomes through the Indian expulsion and not through other time-intervening shocks. Figure 6 examines shocks after 1972 that could independently affect agriculture: later road construction, the return of Indians in the 1990s, out-migration, conflict, and the hosting of refugees.

Panel A of Figure 6 shows that realized Indian market access in 1971 systematically predicts the likelihood of having a road upgraded within 10 kilometers, the likelihood of receiving Indian returnees, and higher share of households with a member moved out. Panel B of Figure 6 shows that the recentered Indian market access does not predict any of the subsequent time-intervening shocks considered. These results rule out these time-intervening shocks as explanations to the main effects.

[12] In Appendix Figure A5, I show that results are not sensitive to the choice of distance thresholds.

Figure 5: Indian Expulsion and Pre-determined Characteristics



Notes: The unit of analysis is the UNPS locality. Each marker is from a separate bivariate OLS regression of Indian exposure on each of the pre-determined characteristic. The dependent variable is realized Indian market access in Panel A and recentered Indian market access (the difference between realized Indian market access and expected Indian market access) in Panel B. Thick bars are 90 percent and thin bars are 95 percent confidence intervals. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999).

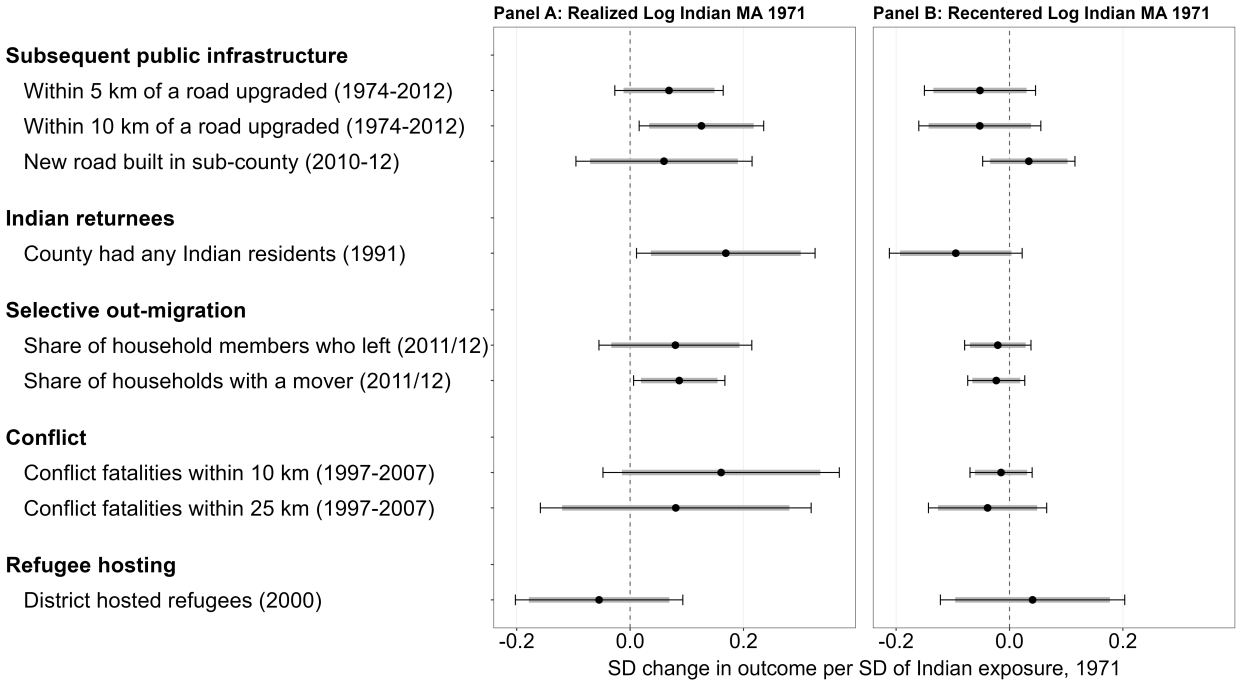
5 Main Results

5.1 Lower Crop Value of Farmland

Did the expulsion lower the value that farmers obtain from their land in the long-run? I test whether households in localities that were better connected to Indian-populated towns in 1971, because of the timing of road upgrades, obtain less crop value from their farmland four decades later. Table 1 reports the estimates of Equation 2 where the outcome is the log crop value of farmland. All regressions control for expected Indian market access. Column 1 includes no other controls, column 2 adds agroclimatic and geographic controls, and column 3 adds household controls.

I find that greater exposure to Indian middlemen in 1971 lowers the crop value of farmland about four decades later. The estimates are negative and significant at the 1 percent level across all specifications. In column 3, a one-standard-deviation increase in recentered exposure lowers the crop value of farmland by 18.9 log points or about 17 percent.

Figure 6: Indian Expulsion and Subsequent Time-intervening Shocks



Notes: The unit of analysis is the UNPS locality. Each marker is the coefficient on log realized Indian market access in 1971 from a separate OLS regression of the outcome indicated in Panel A and recentered log Indian market access (the difference between realized Indian market access and expected Indian market access) in Panel B. Thick bars are 90 percent and thin bars are 95 percent confidence intervals. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999).

5.2 Robustness Checks

I subject the main result to a series of robustness checks, reported in the Appendix. Unless noted otherwise, each check uses the specification of column 3 of Table 1.

Alternative strata. The counterfactual road-completion schedules permute which eligible segments were completed before the expulsion within strata of observably similar segments. In the baseline, the strata are defined by road class interacted with the tercile of a segment's 1961 edge betweenness, the two characteristics on which early and late upgraded road segments differ (see Figure 3). I show that results are similar when strata are defined as the interaction between a segment's road class transition and the median of its 1961 edge betweenness (Appendix Figure A2).

Matching. Coarsened exact matching suggests limited residual bias from differences in geography and pre-expulsion development between more- and less-exposed localities. I split each of the eight agroclimatic and geographic controls at its median and group localities into strata defined by the resulting combinations, dropping strata that do not

Table 1: Indian Expulsion 1972 and Crop Value of Farmland

	(1)	(2)	(3)
Dependent Variable: Log crop value of farmland			
Realized Indian MA 1971	-0.194***	-0.110***	-0.121***
	(0.032)	(0.024)	(0.023)
	[-0.236, -0.133]	[-0.138, -0.073]	[-0.149, -0.084]
Expected Indian MA 1971	0.201***	0.089***	0.104***
	(0.029)	(0.032)	(0.031)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.068	0.123	0.209
Mean of dependent variable	13.447	13.447	13.447
Agroclimatic and geographic controls	–	Y	Y
Household controls	–	–	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is the log-transformed crop value of farmland. The explanatory variable is locality l's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. All regressions exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. Randomization-inference 95% confidence intervals following Borusyak and Hull (2023) are reported in brackets. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

contain both a high- and a low-exposure locality. This leaves 1,095 households in 243 localities across 38 strata.

First, restricting the sample to these matched localities leaves the core findings unchanged (Appendix Figure A3, row 2). Second, including stratum fixed effects, which restricts the comparison to localities that fall on the same side of the median on all eight characteristics, yields similar estimates (Appendix Figure A3, row 3). Even among localities with similar terrain, rainfall, crop suitability, settlement density, and distances to Kampala, and missions, more-exposed households obtain significantly lower crop value of farmland. Third, adding controls of column 3 of Table 1 on top of the stratum fixed effects reduces the estimate but it remains economically and statistically significant (Appendix Figure A3, row 4).

Alternative exposure definitions. The results are robust to alternative measures of exposure (Appendix Figure A4). First, replacing the gravity sum with exposure to the single nearest Indian-populated town, weighted by that town's Indian population, yields an effect nearly identical to the baseline (row 3). Second, the results hold for decay

parameters ranging from 10 to 50 kilometers (rows 4-7). The estimated effect grows modestly as the decay parameter increases.

Alternative inference and samples. Results hold if I use 10km, 30km, 100km Conley SE as well as clustering SE at the locality-, sub-county-, and district-level (Appendix Figure A5). The results are robust. Results hold if I drop one region or sub-region at a time (Appendix Figure A6).

Indian versus general market access. The effect could reflect earlier road access to towns rather than exposure to the Indian commercial network. I note that Indian market access is highly correlated with access to towns in general. To distinguish the two, I restrict the sample to households whose nearest 1959 town, under 1971 road network, had a low Indian share of its population. In these localities, the timing of road upgrades influenced access to towns with few Indians. If the effect captures general market access, we should observe significant effects in this sample.

Appendix Table A6 shows that among households whose nearest town was at most 5 or 10 percent Indian, the estimates are small and statistically insignificant (columns 1 and 2). Raising the cutoff to 15 or 20 percent yields estimates that are negative but small and insignificant (columns 3 and 4).

Other later shocks. The decades after the expulsion brought further shocks that could have affected agriculture independently, including later road construction, the return of Indians after 1992, armed conflict, and the hosting of refugees. As shown in Section 4, recentered exposure does not predict these shocks (Figure 6, Panel B).

Two further shocks were concentrated in particular regions. First, Indians expelled from Uganda may have relocated to Kenya and Tanzania, or Indian intermediaries in these countries may have supplied services to Ugandan farmers near the border. Such cross-border substitution would partly offset the loss of Indian middlemen near the border. The results are robust to excluding localities within 10, 20, 30, and 50 kilometers of the Kenyan and Tanzanian borders, where this substitution is likely to be strongest (Appendix Figure A7). Second, the districts of Luwero, Nakaseke, and Nakasongola, known as the Luwero Triangle, were the site of mass population removal during the 1981–1986 Bush War, which displaced about 750,000 people. The results are robust to excluding localities in the Triangle and within 25, 50, and 75 kilometers of it (Appendix Figure A7).

Persistence over time. The UNPS measures agricultural outcomes only in 2011/2012. To examine whether the decline in crop value reflects a persistent change in agricultural production, I use the settlement-level data, in which agricultural activity is measured by

the normalized difference vegetation index (NDVI). I estimate Equation 2 separately for each four-year interval, with NDVI measured in each historical settlement from 1981–1984 to 2005–2008, including agroclimatic and geographic controls.

Figure A8 shows that more exposed historical settlements have lower vegetation in every period. The expulsion thus appears to have had a lasting effect on agricultural production from 1981 through 2008, suggesting a lasting reduction in agricultural production already visible by 1981.[13]

5.3 Alternative Explanations

My results suggest that households in localities more exposed to the loss of Indian middlemen obtain less crop value from their farmland four decades after the expulsion. Here, I address four alternative explanations based on differences in land use, a general contraction of the local economy, a loss of credit and inputs, and selective out-migration.

Differences in land use. A lower crop value of farmland may simply reflect that more-exposed households farm smaller plots or leave more of their land fallow. Table A7 reports estimates for three outcomes: log size of farmland (Panel A), log crop value per cropped acre (Panel B), and log share of farmland that is cropped (Panel C).

The decline is not explained by farming less land or by cropping the land less intensively. In Panel A, the estimates for the size of farmland are small and statistically insignificant in every specification. In Panel C, the estimates for the share of farmland that is cropped are close to zero and insignificant once agroclimatic and geographic controls are included. The decline instead reflects lower crop value per cropped acre. In Panel B, the estimates are negative and significant at the 1 percent level in all specifications. In column 3, a one-standard-deviation increase in recentered exposure lowers crop value per cropped acre by 12.5 log points (0.080×1.565), or about 12 percent. Taken altogether, households in more-exposed localities farm as much land and plant as much of it, but obtain lesser value from what they plant. This can reflect either a shift toward lower-value crops or lower yields of a given crop, which I examine in Section 6.

A general contraction of local economy. The main analysis excludes localities within towns that had an Indian population in 1959, removing the direct effect of the departure of Indian households themselves. Their departure may nonetheless have depressed demand and economic activity in the surrounding places, lowering the returns to farming

[13] I note that NDVI measures the density of vegetation rather than the quantity or value of crops. It is likely sensitive to the composition of crops. I interpret the NDVI estimates as evidence on the direction and persistence of the main effect.

for reasons unrelated to the loss of middlemen.

Two pieces of evidence argue against this explanation. First, if the effects reflected economic spillovers from the towns, they should weaken substantially as localities near them are excluded. Appendix Figure A7, rows 10 through 13 show that the estimates remain negative, significant, and similar in magnitude when excluding localities within 5, 10, 20, and 30 kilometers of these towns. Second, a general contraction would also reduce economic activity outside agriculture. Table A8 shows that exposure has no significant effect on the presence of nighttime lights in 1992, 2000, or 2010. The effects of the expulsion therefore appear to be specific to agriculture.

A loss of credit and inputs. Indian traders provided services not only in intermediation but also before production. They advanced goods and cash to farmers against future harvests. Indian-owned shops were an important source of seed, tools, and other goods. If the expulsion mainly removed access to credit and inputs, it would have raised the cost of producing crops rather than the cost of selling them, lowering yields directly by limiting farmers' ability to invest in equipment and purchase inputs.

Table A10 shows that exposure has no significant effect on the value of farm implements per acre (Panel A), the number of implements used (Panel C), or the use of labor-saving implements such as ploughs, wheelbarrows, and sprayers (Panel B). Table A11 shows no effect on the use of inorganic fertilizer, or growth regulators, and only a marginally significant, economically small effect on insecticide use. The evidence does not support this channel.[14]

Selective out-migration. If the expulsion led more productive households or members to leave more-exposed localities, the households that remain could be less productive for reasons unrelated to disintermediation.

Three checks suggest that selective out-migration does not drive the results. First, I restrict the sample to households whose head has never left the community. Among households whose head has lived in the community since birth, the core findings are unchanged (Appendix Figure A9, Panel A, row 2). Second, results remain robust for households whose head has lived in the community since 1972 (Panel A, row 3). Third, exposure does not predict out-migration from the household itself (Panel B). The estimated effects on whether any member has left permanently, the share and the number of members who have left, and whether any member has moved out of the dwelling

[14]This is consistent with the historical record: Indian credit to farmers took mainly the form of goods advanced against the harvest, and by the late 1960s cooperatives and state lending institutions supplied much of the credit and inputs available to farmers (Young et al., 1981).

are all small and statistically insignificant. These results suggest that the decline in crop value is not explained by differential migration among the households that remain.

Loss of human capital. The expulsion may have disrupted schooling in more-exposed places, children of school age in 1972 may therefore have completed less schooling and more likely to work in agriculture. I examine this channel with a Duflo-style cohort difference-in-differences (Duflo, 2001). Using the individual records of the 1991 census, I estimate the difference-in-differences on cells of district of birth by year of birth for Ugandan-born persons born 1931–1970.

In Figure A11, the left panels plot raw person-weighted means for the districts of birth above and below the median of recentered Indian market access in 1971. The right panels report the coefficients on realized market access interacted with each cohort indicator, with expected market access interacted with each cohort indicator, birth district fixed effects, and birth year fixed effects. The decline in educational attainment appears in cohorts that were of primary-school age or younger in 1972. However, Figure A12 shows that the loss of educational attainment by exposed cohorts does not translate into changes in employment, suggesting that the loss of human capital is unlikely a channel.

6 Potential Mechanism

The conceptual framework in Section 2.3 attributes the effect of the expulsion to its impact on intermediation capacity: the buyers, processors, transporters, and wholesalers that connect farmers to markets. In localities that depended on Indians for these services, the expulsion left less capacity behind. The main results are consistent with Prediction 1: greater pre-expulsion exposure lowers the crop value of farmland today, implying that the dependence channel dominates the replacement channel. This section tests the remaining predictions, each of which follows from persistent disintermediation.

6.1 Effects are Driven by the Most Marketed Crops

Prediction 2 states that the effect of exposure should be largest for the crops that rely most on intermediation. Before a crop can be sold beyond the village, it typically passes through a sequence of intermediation services: small quantities must be aggregated from many farmers, dried, graded, or processed, stored until they can be sold, and transported to urban or export markets. Crops differ in how much they rely on these services before they become valuable in the market.

To capture this, I rank crops by their total sales value in the 2011/12 UNPS, with each crop valued at its national median price and use this ranking as a proxy for a crop's reliance on intermediation. The five most marketed crops heavily depend on intermediation. For coffee, dried cherries must be hulled, graded, and bulked into export-sized lots before they can be sold to exporters. Maize and beans are grown in small quantities by many farmers and become valuable to urban and regional buyers only once they are aggregated, dried, stored, and transported. Food bananas are bulky and highly perishable, so their value in urban markets depends on collecting bunches from many farms and transporting them quickly to towns. Sugarcane must be milled and processed in sugar factories.

I group crops into nested baskets ordered by sales value: the top 3 (coffee, maize, and food bananas), top 5 (adding sugarcane and beans), and top 10 (adding cocoa, rice, groundnuts, cassava, and cotton). I then estimate the impact of Indian expulsion on crop value of farmland for each basket of the most marketed crops. Table 2 regresses crop value of farmland for baskets of the most marketed crops on realized Indian market access in 1971, using Poisson pseudo-maximum likelihood estimates (PPML). This retains households that harvested none of a basket's crops.

I find that greater exposure to Indian middlemen lowers the crop value of every basket and the effect is the largest for the most marketed crops. The effects are negative and significant in all panels and specifications. In column 3 of Panel A, a one-standard-deviation increase in recentered exposure lowers the value of the top three crops by about 18 percent. As lower-ranked crops are added, the effect becomes smaller. The decline in the crop value of farmland appears to be driven disproportionately by the crops that depend most on intermediation.

6.2 Less of the Harvest Reaches the Market

If fewer intermediaries operate, less of what farmers produce should reach the market. Prediction 3 states that if dependence dominates, greater exposure should reduce participation in crop markets, as farmers shift land toward crops consumed at home. Table 3 reports estimates for an indicator for selling any crop (Panel A) and the percentage of the value of the harvest that is sold (Panel B).

Greater exposure modestly reduces how much farmers sell. In Panel A, the estimates are negative but small and only marginally significant in column 3. A one-standard-deviation increase in recentered exposure lowers the probability of selling any crop by

Table 2: Indian Expulsion and Crop Value of Farmland by Crop Basket

	(1)	(2)	(3)
Panel A. Dep. Var.: Crop value of farmland for top three crop			
Realized Indian MA 1971	-0.211***	-0.119***	-0.130***
	(0.034)	(0.039)	(0.034)
	[-0.383, -0.163]	[-0.159, -0.049]	[-0.175, -0.056]
Expected Indian MA 1971	0.201***	0.098***	0.107***
	(0.038)	(0.032)	(0.027)
Observations	1,811	1,811	1,811
Localities	417	417	417
Pseudo R^2	0.033	0.131	0.174
Mean of dependent variable	0.602	0.602	0.602
Panel B. Dep. Var.: Crop value of farmland for top five crops			
Realized Indian MA 1971	-0.173***	-0.084**	-0.097***
	(0.031)	(0.039)	(0.035)
	[-0.307, -0.144]	[-0.128, -0.037]	[-0.147, -0.046]
Expected Indian MA 1971	0.160***	0.026	0.041
	(0.039)	(0.059)	(0.054)
Observations	1,811	1,811	1,811
Localities	417	417	417
Pseudo R^2	0.022	0.084	0.140
Mean of dependent variable	0.870	0.870	0.870
Panel C. Dep. Var.: Crop value of farmland for top ten crops			
Realized Indian MA 1971	-0.141***	-0.082**	-0.095**
	(0.027)	(0.041)	(0.038)
	[-0.243, -0.113]	[-0.134, -0.044]	[-0.158, -0.052]
Expected Indian MA 1971	0.155***	0.050	0.068
	(0.047)	(0.080)	(0.078)
Observations	1,811	1,811	1,811
Localities	417	417	417
Pseudo R^2	0.014	0.052	0.114
Mean of dependent variable	1.294	1.294	1.294
Agroclimatic and geographic controls	–	Y	Y
Household controls	–	–	Y

Notes: The table reports Poisson pseudo-maximum likelihood estimates of Equation 2, where the dependent variable is the crop value of farmland for the crop basket, in millions of Uganda shillings. Households that harvested none of a basket's crops enter at zero. Baskets are the highest-ranked crops by observed sales value: Panel A is Coffee All, Maize, Banana Food; Panel B adds Sugarcane, Beans; Panel C adds Cocoa, Rice, Groundnuts, Cassava, Cotton. The explanatory variable is locality l's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. All regressions exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. Randomization-inference 95% confidence intervals following Borusyak and Hull (2023) are reported in brackets; for these Poisson estimates, the bracket is the part of the confidence set that contains the estimate. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table 3: Indian Expulsion and Agricultural Commercialization

	(1)	(2)	(3)
Panel A. Dep. Var.: Sells any crop (×100)			
Realized Indian MA 1971	-4.485***	-1.903	-2.008*
	(1.735)	(1.227)	(1.183)
	[-7.058, -0.299]	[-3.822, 0.075]	[-3.892, -0.055]
Expected Indian MA 1971	4.675***	1.062	1.236
	(1.747)	(1.322)	(1.279)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.036	0.073	0.084
Mean of dependent variable	80.453	80.453	80.453
Panel B. Dep. Var.: Percentage of harvest value sold			
Realized Indian MA 1971	-2.032***	-1.019*	-1.099**
	(0.576)	(0.534)	(0.501)
	[-3.002, -0.495]	[-1.954, 0.000]	[-1.961, -0.159]
Expected Indian MA 1971	2.125***	0.862	1.021*
	(0.569)	(0.597)	(0.568)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.018	0.043	0.074
Mean of dependent variable	28.617	28.617	28.617
Agroclimatic and geographic controls	–	Y	Y
Household controls	–	–	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is an indicator for selling any crop output in Panel A and the percentage of the value of the household's harvest that was sold in Panel B. Both are scaled by 100, so coefficients are percentage points. Households selling nothing enter Panel B at zero. The explanatory variable is locality l's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. All regressions exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. Randomization-inference 95% confidence intervals following Borusyak and Hull (2023) are reported in brackets. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

3.1 percentage points, against a mean of 80 percent. In Panel B, the same increase lowers the share of the harvest sold by 1.7 percentage points, against a mean of 29 percent, an effect significant at the 5 percent level. A small share of households stops selling crops, and the share of the harvest that reaches the market falls, consistent with a wider wedge between the market price and what farmers receive at the farm gate.

6.3 Sales Shift Away from Traders

If farmers shifted land toward crops sold locally, Prediction 4 implies that their sales should shift away from traders toward direct sales to consumers. I test this using the survey's report of the buyer of the largest part of each sale and test whether households in more exposed localities are less likely to sell to private traders and more likely to sell directly to consumers, conditional on households that sell. In the survey, traders are private traders in the local village or market and in the district market and non-traders are consumers at a market, neighbors and relatives, the government, and other buyers.

Table 4 shows that sales in more-exposed localities shift away from traders. In column 3, a one-standard-deviation increase in recentered Indian market access reduces the probability of selling any harvest to a trader by 4.7 percentage points (Panel A) and increases the probability of selling any output to a non-trader by 4.2 percentage points (Panel B). Moreover, the share of sales value going to traders falls by 6.2 percentage points (Panel C), against a mean of 75 percent. Altogether, these results suggest that in more exposed localities, crops reach buyers through shorter, more local channels, which are less suited to the crops that generate the most value.

6.4 Taking Stock

Together, these results point to persistent disintermediation as the mechanism behind the lower crop value of farmland. First, the decline is largest for the crops that depend most on intermediation. Second, fewer households in more-exposed localities sell crops at all and a smaller share of the harvest reaches the market. Third, among households that sell, sales shift away from traders toward consumers, neighbors, and other buyers. These patterns are difficult to reconcile with the replacement channel, under which more-exposed localities would have rebuilt the intermediation and shown the opposite patterns.

7 Downstream Impact on Consumption

So far, I have shown that more-exposed households obtain less value from their farmland. A natural question is whether this lower crop value translates into lower household welfare.

Table 4: Indian Expulsion and the Composition of Crop Buyers

	(1)	(2)	(3)
Panel A. Dep. Var.: Sells any output to a trader (×100)			
Realized Indian MA 1971	-3.216***	-2.880***	-2.987***
	(0.772)	(0.555)	(0.563)
	[-4.260, -2.133]	[-3.737, -1.779]	[-3.844, -1.853]
Expected Indian MA 1971	3.746***	2.841***	3.008***
	(0.850)	(0.682)	(0.709)
Observations	1,454	1,454	1,454
Localities	368	368	368
R^2	0.029	0.056	0.070
Mean of dependent variable	87.208	87.208	87.208
Panel B. Dep. Var.: Sells any output to a non-trader (×100)			
Realized Indian MA 1971	1.911**	2.621***	2.695***
	(0.757)	(0.688)	(0.672)
	[0.533, 3.529]	[1.260, 4.089]	[1.335, 4.140]
Expected Indian MA 1971	-2.634***	-3.101***	-3.191***
	(0.822)	(0.879)	(0.866)
Observations	1,454	1,454	1,454
Localities	368	368	368
R^2	0.011	0.029	0.031
Mean of dependent variable	48.693	48.693	48.693
Panel C. Dep. Var.: Share of sales value sold to traders (×100)			
Realized Indian MA 1971	-3.561***	-3.839***	-3.941***
	(0.531)	(0.521)	(0.520)
	[-4.609, -2.513]	[-4.870, -2.565]	[-4.981, -2.652]
Expected Indian MA 1971	4.182***	4.217***	4.384***
	(0.574)	(0.586)	(0.597)
Observations	1,454	1,454	1,454
Localities	368	368	368
R^2	0.031	0.047	0.060
Mean of dependent variable	75.082	75.082	75.082
Agroclimatic and geographic controls	–	Y	Y
Household controls	–	–	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is an indicator for selling any crop output to a trader in Panel A, an indicator for selling any output to a non-trader in Panel B, and the percentage of the household's crop sales value sold to traders in Panel C; all are scaled by 100. Traders are private traders in the local village or market and in the district market; non-traders are consumers at market, neighbours and relatives, government and other buyers. The explanatory variable is locality l's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. All regressions exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. Randomization-inference 95% confidence intervals following Borusyak and Hull (2023) are reported in brackets. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

7.1 Impact on household consumption

I use the consumption aggregates constructed by the Uganda Bureau of Statistics for the 2011/2012 UNPS, expressed as monthly consumption per adult equivalent and deflated to 2005/2006 prices. I examine three components: food consumption covers food, beverages, and tobacco over the past seven days, scaled to thirty days, and includes food produced by the households; non-durable non-food consumption covers spending over the past thirty days on housing, fuels and utilities, personal and household supplies, transport, communication, health care, and personal services; durable consumption covers spending over the past 365 days on education, clothing and footwear, household durables, and insurance, scaled to thirty days.

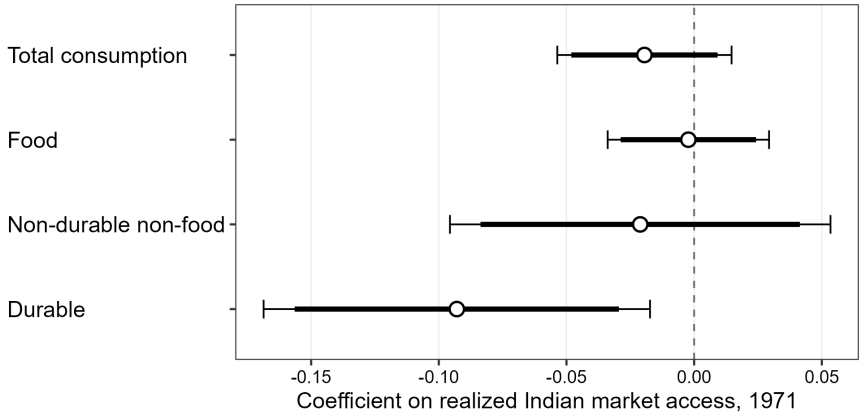
Figure 7 shows estimates using Poisson pseudo-maximum likelihood, with each consumption measure as the outcome. All regressions include expected Indian market access, agroclimatic and geographic controls as well as household controls. Lower crop value translates into lower spending on durable goods. A one-standard-deviation increase in recentered exposure significantly lowers durable spending by about 13 percent at the 5 percent level and close in magnitude to the decline in crop value documented above. In contrast, the effects on food consumption and total consumption are small and statistically insignificant. This pattern is consistent with households maintaining their food consumption through home production, since they sell a smaller share of their harvest, while absorbing the loss in crop value by cutting back on durable purchases.

8 Discussion and Conclusion

Ethnic minorities often dominate particular occupations in their host economies, yet little is known about whether others can take their place once the group is gone. This paper studies one of the starkest removals of such a group–the 1972 expulsion of Uganda's Indian minority, which had occupied the intermediary layer of the country's agricultural value chain. Using newly digitized census and road network data, I measure each locality's exposure to the Indian commercial network on the eve of the expulsion and isolate variation in exposure from which segments of a road upgrading program happened to be completed before 1972.

Four decades later, households in more-exposed localities obtain substantially lower crop value from their farmland. They farm as much land and plant as much of it as other households, but obtain less value from what they plant. Satellite data show lower

Figure 7: Indian Expulsion and Household Consumption



Notes: The figure reports Poisson pseudo-maximum-likelihood estimates of the coefficient on realized log Indian market access in 1971 for monthly consumption per adult equivalent, deflated to 2005/2006 prices. Total consumption is the consumption aggregate; food is food, beverages, tobacco over the past 7 days, scaled to 30 days; non-durable non-food is spending over the past 30 days on housing, fuels and utilities, personal and household supplies, transport, communication, health care, and personal services; durable is spending over the past 365 days, scaled to 30 days, on education, clothing and footwear, household durables, insurance. All regressions include expected log Indian market access in 1971, the number of populated places within 10km in 1964, ruggedness, rainfed cotton and coffee suitability, average annual rainfall over 1960–1969, and distances to the nearest 1959 town, to Kampala, and to the nearest mission. Thick bars are 90 per cent and thin bars 95 per cent confidence intervals, constructed from standard errors allowing for spatial correlation among observations within 50 km following Conley (1999).

vegetation in more-exposed places from 1981 onward. The evidence points to persistent disintermediation. The decline is largest for the crops that depend most on intermediation services; more-exposed households sell less of their harvest; and those who sell rely less on traders and more on consumers, neighbours, and other local buyers. The effect is specific to exposure to Indian commerce rather than to towns in general, and it is not explained by a loss of credit and inputs, a general contraction of the local economy, selective out-migration, or later shocks correlated with exposure.

These findings suggest that the niche the Indians occupied rested on group-specific network capital that could not simply be inherited. Their successors took over the shops, lorries, and mills, but not the chains of credit and trust that had linked farmers to distant markets. The expulsion was justified as an “economic war" to return commerce to African hands. In the localities most dependent on Indian commerce, its long-run effect was instead to leave farmers with fewer intermediaries and less value from their farmland.

The results have implications beyond Uganda. Governments across the world re-

strict, tax, or displace traders on the grounds that they capture margins that would otherwise accrue to farmers. When those traders belong to a minority whose commercial advantage rests on networks rather than on transferable skills, such policies may not redistribute those margins but eliminate the services that generated them. More broadly, the findings suggest that the gains from exchange between ethnic groups that specialize in different activities may be fragile. Once it is interrupted, they need not be rebuilt by the groups that remain.

There are several open questions. First, the analysis shows that the intermediaries were not replaced, but not *why* entry remained low over four decades. Understanding what prevented the rebuild, whether it was limited commercial skills, barriers to entry, would clarify when ethnic niches can be refilled. Second, similar expulsions of commercial minorities, from the Chinese in Southeast Asia to traders in West Africa, offer settings in which to test whether the pattern documented here is general.

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APPENDIX

A Data and Summary Statistics

This appendix describes the data used in the analysis and how I construct each variable. The analysis combines five sources: the 1959 census of Uganda's non-African population, historical road maps recording the surface class of each road segment, the 2011/12 wave of the Uganda National Panel Survey, the 1964 gazetteer of Ugandan populated places, and gridded geographic and satellite data. I first describe the sources and then detail the construction of Indian market access, and of the counterfactual road networks that define the expected exposure. Next I turn to household outcomes, the sample restrictions, the controls and the settlement-level data.

A.1 Data Sources

1959 census of the non-African population. The East African Statistical Department enumerated Uganda's non-African population separately from the African population in 1959 (East African Statistical Department, 1960). Its town tables report the population of each town and trading centre by community. I define the Indian population of a town as the sum of the census categories "Indian", "Pakistani", and "Goan". This is the same aggregation the 1969 census uses for its Asian total.

I geocode each town by matching it to its entry in the 1964 gazetteer of Uganda (described below). I match on the town's name within its district, using the gazetteer's area code, which identifies the district in which a place lies. For cases where names differ only in spelling or spacing (for example, "Aboke" and "Aboki"), I match by hand. I assign each town the gazetteer's latitude and longitude. One town, Lwambu in Masaka district, cannot be located and is dropped.

1964 gazetteer. The gazetteer is Office of Geography (1964), which lists every named geographic feature in the country, including populated places, missions, hills and streams. For each feature, it gives the approved name, a feature designation, latitude and longitude (in degrees and minutes), and an area code identifying the district.

I digitize the gazetteer from a scanned copy. I extract the text of each page and parse each entry into these fields. The digitized file contains 11,198 entries, of which 9,334 are main entries. I keep the 4,756 entries designated as populated places. Of these, 3,994

have both coodinates recovered and the rest lost a coordinate in digitization and are dropped. I use the 3,881 populated places as the units of the settlement-level analysis, and counts of them form the settlement-density control.

Road network. The road data come from Jedwab and Storeygard (2022). They digitize the road network of sub-Saharan African countries from Michelin and other road maps and record each segment's surface class in a series of benchmark years. I use the Ugandan network in 1961, 1971, and 1974 and add the Uganda Railway, whose alignment did not change over the period considered.

Uganda National Panel Survey (UNPS), 2011/12. The UNPS is conducted by the Uganda Bureau of Statistics under the World Bank's LSMS-ISA program. The 2011/12 wave is nationally representative. Its agriculture module records parcels, plots, crops, harvests, sales and buyers for the two copping seasons of 2011. Survey enumerators record the GPS location of each dwelling. The public release does not report these locations. Instead, the World Bank publishes modified coordinates that are displaced to protect respondents' confidentiality and are, for most households, common to all households in an enumeration area. I use the released coordinates directly and define a locality as a distinct pair of them.

A.2 Construction of Indian Market Access

Road network. I convert the road and railway lines into a network in which each segment is an edge. Segments are split at every point where they intersect another segment, so that a traveller can switch roads at any junction. A segment without a recorded class is treated as an earth road.

Travel costs. The cost of traversing an edge in year t is its length in kilometers multiplied by a cost factor that depends on its surface class in t. Costs are measured in kilometer-equivalents (See Appendix Table A1).

Each UNPS locality and each 1959 town is assigned to the nearest node of the network. The travel cost $\tau_{lj,t}$ between locality l and town j is the least-cost path through the network in year t, computed with Dijkstra's algorithm. Pairs that the network does not connect are assigned twice the largest finite cost. This affects three localities, which I exclude from the sample.

Table A1: Road Network by Surface Class, 1961 and 1971

		Number of segments	
Surface class	Cost factor	1961	1971
Paved	1.0	135	195
Improved	1.5	185	297
Earth	3.0	1,093	921
Railway	1.0	12	12
Total		1,425	1,425

Notes: Road segments and surface classes are from Jedwab and Storeygard (2022). The cost of traversing a segment is its length in kilometers multiplied by the cost factor of its surface class.

Measure. Indian market access of locality l in 1971 is

$$IndianMA_{l,1971} = \sum_{j=1}^{N} \frac{IndPop_j}{\tau(Network_{1971}, Loc_l, Loc_j)} \tag{3}$$

with distance-decay parameter $\theta = 10$ kilometer-equivalents. All regressions use the natural logarithm of the measure.

Localities inside a town. A locality assigned to the same network node as a 1959 town has zero travel cost to that town. Its exposure therefore reflects the presence of the Indian population itself rather than access to it. I classify such localities as lying inside a 1959 town and exclude them.

A.3 Counterfactual Road Networks and Expected Exposure

Eligible segments. A segment is eligible if its surface class improved between 1961 and 1974. There are 291 segments in total. A segment is *completed* if it had reached its 1974 class by 1971. That is the case for 222 segments. The other 69 were completed between 1972 and 1974.

Strata. Completed segments were more central in the 1961 network than delayed ones and they differ in their class transition (See Figure 3). I therefore draw counterfactual completions within strata. A stratum is defined as the interaction of two characteristics: (i) the segment's class transition; (ii) the tercile of its edge betweenness in the 1961 road network. Edge betweenness is the number of least-cost paths between pairs of nodes that pass through the segment, computed with 1961 travel costs and summed over the edges into which the segment is split. 22 segments lie in strata in which every segment

was completed, hence their status never varies across draws.

Draws. In each draw I select, within every stratum, as many segments as were actually completed there. The counterfactual 1971 network has three components: the drawn segments at their 1974 class, the remaining eligible segments at their 1961 class, and every ineligible segment at its actual 1971 class. I recompute Indian market access for every locality on each of 1,000 draws, using a fixed random seed. Expected exposure is the mean of log Indian market access across the 1,000 draws.

A.4 Household Outcomes

All household outcomes come from the UNPS agriculture module.

Farmland. I build farmland from the parcel rosters for owned parcels (AGSEC2A) and rented-in parcels (AGSEC2B). A parcel counts as farmland if two conditions hold: (i) its primary use in either season is own cultivation of annual or perennial crops or fallow; (ii) the household currently has access to it. Parcels that are rented out, sub-contracted out, or used as woodlot, forest or pasture are excluded. This leaves 96.5 percent of parcels and 94.2 percent of parcel area. Parcel area is the GPS measurement where available and the farmer's estimate otherwise.

Crop value of farmland. Harvests come from the crop-by-plot-by-season rosters (AGSEC5A and AGSEC5B). The construction has three steps:

1. *Quantities in kilograms.* I convert each harvest into kilograms using the unit-specific conversion factors supplied by UBoS. Harvests reported directly in kilograms carry no conversion factor.

2. *Prices.* I value each crop at its national median sale price. Since every household faces the same price for a given crop, differences in crop value reflect differences in the quantities and composition of what households harvest, instead of local price differences.

3. Crop value is the sum of harvest value over all crops, plots, and both seasons of 2011.

The estimation sample covers 43 crops. Its median crop value is 735,385 Uganda shillings. Crop rows whose harvest unit has no conversion factor contribute zero, so crop value is a lower bound for the households that report them.

Cropped area. Cropped area is built from the same parcel roster as farmland. In each season, a farmland parcel contributes its area multiplied by the reported share planted if its primary use that season is own cultivation, and zero otherwise. The share planted is asked only for owned parcels. Rented-in parcels are treated as fully planted.

Crop baskets. I rank crops by their total sales value in the survey (See Appendix Table A5). The five largest are coffee, maize, food bananas, sugarcane and beans, which together account for about 60 percent of sales value. For each nested basket (top 3, top 5, top 10), crop value is the median-price value of the harvest of the crops in the basket.

Market participation and buyers. A household participates in crop markets if it sold any crop in either season. The share of harvest sold is the value of sales divided by the value of the harvest, both at median prices.

For each crop sold, the survey records only the buyer of the largest part of the sale. I classify buyers into two groups: (i) traders: private traders in the village or in a district market; (ii) non-traders: consumers at market, neighbours and relatives, government, and other buyers. Since only the main buyer is recorded, I assign the entire sale of a crop to that buyer's group.

Consumption. I use the consumption aggregates published by UBoS. They are expressed as monthly consumption per adult equivalent and deflated spatially and temporally to 2005/06 prices.

Household controls. These are the gender and age of the household head and the number of household members, taken from the household roster.

A.5 Data Processing and Sample Construction

I construct the estimation sample in five steps. First, the starting point is the 2,169 households that complete the agriculture module and have released coordinates. They live in 488 localities. I drop 21 households in 3 localities that the 1971 network does not connect to any 1959 town, because their exposure is undefined. I then drop 251 households in 58 localities that lie inside a 1959 town. I drop 64 households with a missing value on any control in the most saturated specification, leaving 1,833 households in 419 localities. I keep households with positive farmland and positive crop value, which drop 8 and 14 households respectively.

A.6 Controls

Terrain ruggedness. This is the World Bank's terrain-roughness geovariable distributed with the UNPS, computed from SRTM elevation data.

Cotton and coffee suitability. The data are from the FAO Global Agro-Ecological Zones (GAEZ) suitability indices for high-input, rain-fed cultivation.

Rainfall. It is mean annual precipitation over 1960–1969, from the WorldClim historical monthly weather series. For each year, I sum the monthly totals and then average over the ten years.

Settlement density. It is the number of populated places in the 1965 gazetteer within 10 kilometers of the locality.

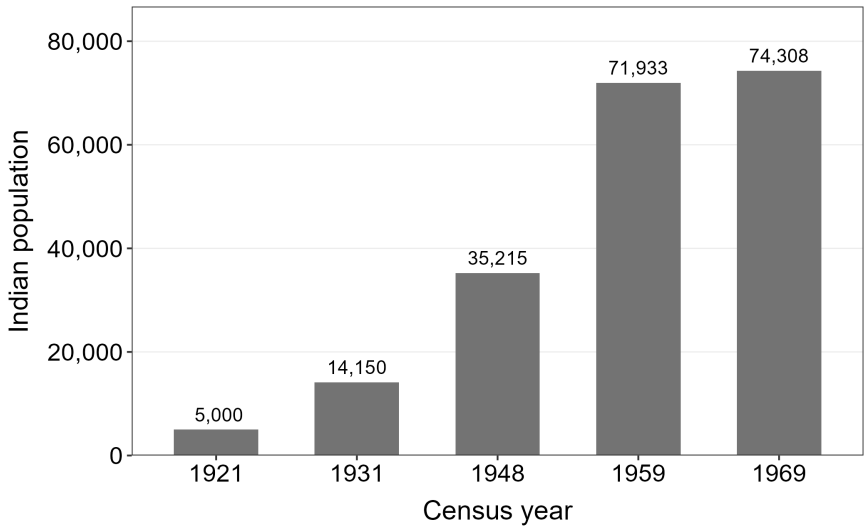
Distance to the nearest 1959 town. This is the straight-line distance to the nearest town with an Indian population in the 1959 census.

Distance to Kampala. This is the straight-line distance to Kampala.

Distance to the nearest mission. This is the straight-line distance to the nearest Christian mission station in Nunn (2010).

B Additional Figures and Tables

Figure A1: Indian population of Uganda, 1921–1969



Notes: Figures for 1921, 1931, 1948, and 1959 are from the protectorate censuses, as reported in Hundle (2013). The 1948 and 1959 figures are from the non-native censuses, which enumerated non-Africans separately. The 1969 figure is from the 1969 census (Uganda. Ministry of Planning and Economic Development. Statistics Division, 1971). The 1921 figure is rounded in the source.

Table A2: Indian and Total Population of Ugandan Towns, 1959

Town	District	Indian	Total	Town	District	Indian	Total
Aboke	Lango	63	72†	Kisoro	Kigezi	63	68†
Aduku	Lango	90	91†	Kitgum	Acholi	199	3,454
Aloi	Lango	37	37†	Kumi	Teso	219	1,037
Amuria	Teso	41	711	Kyenjojo	Toro	48	60†
Arua	West Nile	516	4,645	Kyere	Teso	96	106†
Atura	Lango	71	71†	Lira	Lango	671	2,929
Bombo	East Mengo	410	432†	Lugazi	East Mengo	305	8,105
Budadiri	Bugisu	41	42†	Majanji	Bukedi	47	47†
Budaka	Bukedi	18	68†	Masaka	Masaka	2,139	4,782
Bududa	Bugisu	104	104†	Masindi	Bunyoro	401	1,571
Bugiri	Busoga	275	392†	Masindi Port	Bunyoro	81	601
Buikwe	East Mengo	188	231†	Mayenze	Bugisu	80	80†
Bukedea	Teso	71	102†	Mazimasa	Bukedi	0	10†
Bulopa	Busoga	255	271†	Mbale	Mbale	4,575	13,569
Bundibugyo	Toro	18	1,615	Mbarara	Ankole	919	3,844
Busembatia	Busoga	264	1,316	Mbulamuti	Busoga	67	67†
Bushenyi	Ankole	136	146†	Minakulu	Lango	57	60†
Busia	Bukedi	103	108†	Mityana	West Mengo	264	291†
Busiu	Bugisu	22	31†	Molo	Bukedi	84	89†
Busolwe	Bukedi	81	84†	Moroto	Karamoja	77	2,082
Butaleja	Bukedi	23	23†	Moyo	Madi	37	2,009
Butiaba	Bunyoro	49	1,216	Mpigi	West Mengo	187	191†
Butiru	Bugisu	52	52†	Mubende	Mubende	102	1,877
Butiti	Toro	14	33†	Mukono	East Mengo	208	224†
Entebbe	West Mengo	904	10,941	Muntemo	Bunyoro	17	17†
Fort Portal	Toro	519	8,317	Naboa	Bukedi	17	17†
Gulu	Acholi	571	4,770	Nagalama	East Mengo	125	180†
Hoima	Bunyoro	312	1,056	Nagongera	Bukedi	181	222†
Iganga	Busoga	1,005	3,146	Naigobia	Busoga	49	74†
Ishasha	Kigezi	6	6†	Namasagali	Busoga	100	100†
Jinja	Busoga	8,883	29,741	Namirembe	West Mengo	422	429†
Kabale	Kigezi	611	10,919	Namutumba	Busoga	77	118†
Kabataro	Toro	74	80†	Namwendwa	Busoga	117	149†
Kaberamaido	Teso	314	342†	Namwiwa	Busoga	22	97†
Kakoro	Bukedi	25	33†	Nawaikoke	Busoga	27	63†
Kalaki	Teso	60	60†	Ngora	Teso	175	233†
Kaliro	Busoga	253	914	Njeru	East Mengo	8	1,988
Kalisizo	Masaka	173	759	Otuboi	Teso	121	130†
Kampala	West Mengo	19,268	46,735	Pallisa	Bukedi	258	292†
Kamuli	Busoga	817	1,867	Petete	Bukedi	32	32†
Kasese	Toro	46	1,564	Port Bell	West Mengo	33	102†
Katakwi	Teso	79	661	Serere	Teso	196	1,181
Katunguru	Ankole	5	6†	Siroko	Bugisu	38	52†
Kawempe	West Mengo	401	436†	Soroti	Teso	1,833	6,645
Kibuku	Bukedi	24	49†	Tororo	Bukedi	1,220	6,365
Kidera	Busoga	60	60†	Vukula	Busoga	16	67†
Kikagati	Ankole	15	17†	Wobulenzi	East Mengo	275	282†
				All towns		53,652	200,360

Notes: Town-level counts from the 1959 census of Uganda (94 towns). Indian population includes Pakistanis and Goans. Total population sums all recorded groups. African population is not reported for 60 of 94 towns and is coded as zero (†), so their totals understate the town population.

Table A3: Summary Statistics, Household Level

	Obs.	Mean	St. Dev.	Min.	Max.
Indian market access, 1971:					
Realized log Indian market access	1,811	-0.871	6.659	-31.799	9.535
Expected log Indian market access	1,811	-0.885	6.976	-36.650	9.535
Recentered log Indian market access	1,811	0.014	1.565	-8.240	6.714
Crop value of farmland:					
Crop value of farmland (millions UGX)	1,811	1.709	7.471	0.004	192.994
Log crop value of farmland	1,811	13.447	1.231	8.412	19.078
Decomposition of crop value:					
Farmland (acres)	1,811	4.286	7.287	0.010	100.000
Log farmland	1,811	0.916	1.008	-4.605	4.605
Crop value per cropped acre (millions UGX)	1,811	0.863	3.845	0.001	111.830
Log crop value per cropped acre	1,811	12.933	1.033	7.131	18.532
Cropped area (acres)	1,811	2.531	3.289	0.008	70.000
Cropped area as a share of farmland	1,811	0.741	0.263	0.005	1.000
Log cropped area as a share of farmland	1,811	-0.402	0.540	-5.298	0.000
Crop value per acre of farmland, by basket:					
Top three crops (millions UGX)	1,515	0.315	2.905	0.000	111.505
Log, top three crops	1,473	11.411	1.548	6.215	18.530
Top five crops (millions UGX)	1,657	0.378	2.903	0.000	111.673
Log, top five crops	1,620	11.678	1.515	6.215	18.531
Top ten crops (millions UGX)	1,782	0.470	2.957	0.000	111.830
Log, top ten crops	1,748	12.109	1.347	6.215	18.532
Crop value of farmland, by basket including non-producers (millions UGX):					
Top three crops	1,811	0.602	4.194	0.000	167.257
Top five crops	1,811	0.870	5.284	0.000	167.509
Top ten crops	1,811	1.294	7.161	0.000	192.640
Capital and implements:					
Implement value per acre of farmland (millions UGX)	1,805	0.024	0.125	0.000	4.777
Log implement value per acre of farmland	1,800	9.208	1.135	3.743	15.379
Uses any labour-saving implement (×100)	1,805	27.535	44.681	0.000	100.000
Number of implements used	1,805	3.789	2.695	0.000	54.000
Log number of implements used	1,790	1.163	0.595	0.000	3.989
Crop marketing:					
Sells any crop (×100)	1,811	80.453	39.667	0.000	100.000
Percentage of harvest value sold	1,811	28.617	25.911	0.000	100.000
Sells any output to a trader (×100)	1,454	87.208	33.412	0.000	100.000
Sells any output to a non-trader (×100)	1,454	48.693	50.000	0.000	100.000
Share of sales value sold to traders (×100)	1,454	75.082	36.458	0.000	100.000
Purchased inputs:					
Uses inorganic fertiliser (×100)	1,811	5.743	23.272	0.000	100.000
Uses insecticide (×100)	1,811	8.614	28.065	0.000	100.000
Uses a growth regulator (×100)	1,811	5.356	22.521	0.000	100.000
Agroclimatic and geographic controls:					
Ruggedness	1,811	6.939	3.357	3.000	14.000
Cotton suitability (rainfed)	1,811	5105.496	2502.762	0.000	10000.000
Coffee suitability (rainfed)	1,811	5578.192	2105.813	0.000	9344.000
Average annual rainfall, 1960–1969 (mm)	1,811	1298.103	212.276	750.730	1952.690
Populated places within 10 km, 1964	1,811	9	6	0	28
Distance to nearest 1959 town (km)	1,811	25.527	22.689	0.725	138.393
Distance to Kampala (km)	1,811	209.849	94.482	4.721	403.911
Distance to nearest mission (km)	1,811	39.108	38.011	0.033	219.066
Household controls:					
Household head is female	1,811	0.304	0.460	0.000	1.000
Age of household head	1,811	47.909	14.944	16.000	97.000
Household size	1,811	7.948	3.816	1.000	33.000

Table A4: Summary Statistics, Settlement Level

	Obs.	Mean	St. Dev.	Min.	Max.
Indian market access, 1971:					
Realized log Indian market access	3,429	-0.884	6.018	-39.089	9.797
Expected log Indian market access	3,429	-0.962	6.208	-42.767	9.797
Recentered log Indian market access	3,429	0.079	1.047	-6.824	6.714
Night-time lights, 1992:					
Lit ($\times$100)	3,429	4.899	21.589	0.000	100.000
Mean digital number within 5 km	3,429	0.156	1.622	0.000	40.849
Log mean digital number, lit settlements	168	-0.073	1.574	-2.923	3.710
Night-time lights, 2000:					
Lit ($\times$100)	3,429	13.386	34.055	0.000	100.000
Mean digital number within 5 km	3,429	0.365	2.512	0.000	53.199
Log mean digital number, lit settlements	459	-0.433	1.782	-3.839	3.974
Night-time lights, 2010:					
Lit ($\times$100)	3,429	15.165	35.873	0.000	100.000
Mean digital number within 5 km	3,429	0.627	3.565	0.000	59.966
Log mean digital number, lit settlements	520	0.307	1.508	-2.686	4.094
Agroclimatic and geographic controls:					
Ruggedness	3,429	77913.227	94973.381	0.000	943146.000
Cotton suitability (rainfed)	3,429	5494.357	2437.190	0.000	10000.000
Coffee suitability (rainfed)	3,429	5870.829	1954.353	0.000	9765.000
Average annual rainfall, 1960–1969 (mm)	3,429	1293.872	201.997	707.720	2288.140
Populated places within 10 km, 1964	3,429	11	6	0	33
Distance to nearest 1959 town (km)	3,429	25.397	18.151	1.842	144.808
Distance to Kampala (km)	3,429	174.837	93.088	1.782	420.848
Distance to nearest mission (km)	3,429	37.294	31.097	0.348	235.358

Table A5: Crop Sale Value in the Uganda National Panel Survey, 2011–2012

Crop	Value (m UGX)	% of National	Crop	Value (m UGX)	% of National
Coffee All	262.7	18.59	Banana Beer	10.8	0.76
Maize	179.3	12.69	Oranges	8.3	0.59
Banana Food	167.5	11.85	Tea	8.2	0.58
Sugarcane	148.8	10.53	Onions	8.2	0.58
Beans	89.5	6.33	Banana Sweet	6.8	0.48
Cocoa	78.5	5.56	Field Peas	3.0	0.21
Rice	55.0	3.89	Passion Fruit	2.8	0.20
Groundnuts	49.6	3.51	Yam	2.7	0.19
Cassava	47.8	3.38	Pumpkins	2.7	0.19
Cotton	45.1	3.19	Vanilla	2.2	0.15
Sun Flower	36.0	2.55	Avocado	1.7	0.12
Tobacco	28.0	1.98	Eggplants	1.3	0.09
Soya Beans	23.1	1.64	Pineapples	1.2	0.09
Sweet Potatoes	20.3	1.44	Jackfruit	0.9	0.06
Other	19.7	1.40	Cow Peas	0.8	0.06
Tomatoes	17.0	1.21	Mango	0.7	0.05
SimSim	16.0	1.13	Pigeon Peas	0.6	0.04
Finger Millet	15.0	1.06	Dodo	0.3	0.02
Cabbage	13.7	0.97	Chick Peas	0.0	0.00
Irish Potatoes	13.6	0.97	Coco Yam	0.0	0.00
Pawpaw	12.5	0.89	Wheat	0.0	0.00
Sorghum	11.1	0.79			
			Total	1,413.5	100.00

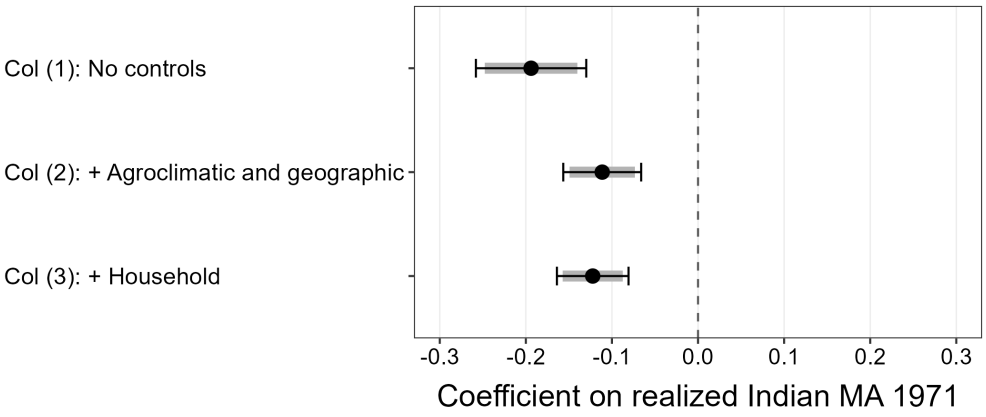
Notes: The table reports the total value of crop sales by crop, and each crop's share of total crop-sale value, in the 2011–2012 wave of the Uganda National Panel Survey. Crops are sorted by sale value, reading down the left block and then down the right. Sale value is the sum of reported sale value over plot-crop-season records, pooling both seasons, and is gross of transport costs. Records reporting no sale value contribute zero. Sales are highly concentrated: the five largest crops account for 60.0 percent of all crop-sale value and the ten largest for 79.5 percent. *Other* is a survey response category rather than a residual computed here. The Uganda National Panel Survey carries no sampling weights, so the figures are unweighted sample aggregates and not population estimates.

C Robustness of the Main Result

This appendix reports the robustness checks summarized in Section 5. Unless noted otherwise, every estimate uses the specification of column 3 of Table 1, which controls for expected Indian market access and the agroclimatic, geographic, and household controls, and excludes localities within 1959 towns.

Alternative strata. The counterfactual road-completion schedules permute which eligible segments were completed by 1971 within strata. The baseline strata interact a segment's road-class transition with the tercile of its 1961 edge betweenness. Appendix Figure A2 replaces the terciles with a median split, which yields six strata instead of nine. I redraw 1,000 schedules and recompute expected exposure from them. The estimates are virtually unchanged to the baseline in all three specifications.

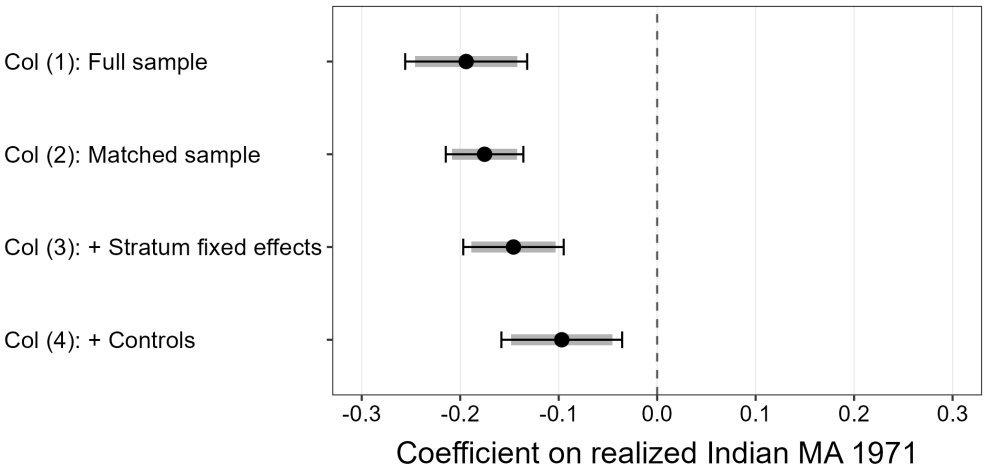
Figure A2: Road-Class Transition Interacted with Betweenness Median



Notes: The figure reports the coefficient on realized Indian market access in 1971 from the specifications of Table 1, when the counterfactual road-completion schedules permute completion within strata defined by a segment's road-class transition interacted with the median of its 1961 edge betweenness, rather than the baseline strata of road class interacted with betweenness terciles. Thick and thin bars are 90 and 95 percent confidence intervals, based on standard errors allowing for spatial correlation among observations within 50 kilometers following Conley (1999).

Coarsened exact matching. Appendix Figure A3 restricts the comparison to localities with similar geography. I split each of the eight agroclimatic and geographic controls at its median across localities and define a stratum as one combination of the resulting bins. Of the 256 possible strata, I keep the 38 that contain both a locality above and a locality below median recentered exposure, leaving 1,095 households in 243 localities. Across all four specifications, the estimates are significant at the 1 percent level.

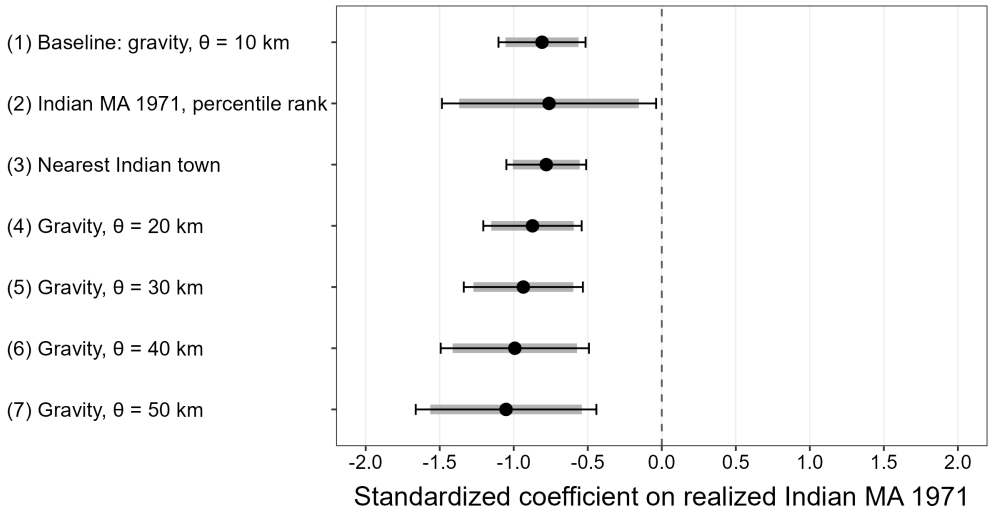
Figure A3: Coarsened Exact Matching



Notes: The figure reports the coefficient on realized Indian market access in 1971 from the specification of column 3 of Table 1. Row 1 is the full sample. Rows 2 to 4 restrict it to a coarsened exact matching sample: each of the eight agroclimatic and geographic controls is split at its median across localities, a stratum is a combination of these bins, and strata without both a high- and a low-exposure locality are dropped, leaving 1,095 households in 243 localities and 38 strata. Row 3 adds stratum fixed effects and row 4 adds the controls of column 3 of Table 1. Thick and thin bars are 90 and 95 percent confidence intervals, based on standard errors allowing for spatial correlation among observations within 50 kilometers following Conley (1999).

Alternative exposure measures. Appendix Figure A4 varies the construction of exposure while holding the counterfactual schedules fixed. For each alternative measure, I recompute both realized and expected exposure on the same 1,000 schedules. As the measures are on different scales, the figure reports each coefficient multiplied by the standard deviation of realized exposure under that measure. First, replacing log market access with its percentile rank gives an estimate of similar size, though it is less precise. Second, replacing the gravity sum over all towns with exposure to the single nearest town, weighted by its Indian population, yields similar estimate. Third, raising the decay parameter from 10 to 20, 30, 40, and 50 kilometer-equivalents gives similar estimates to the baseline.

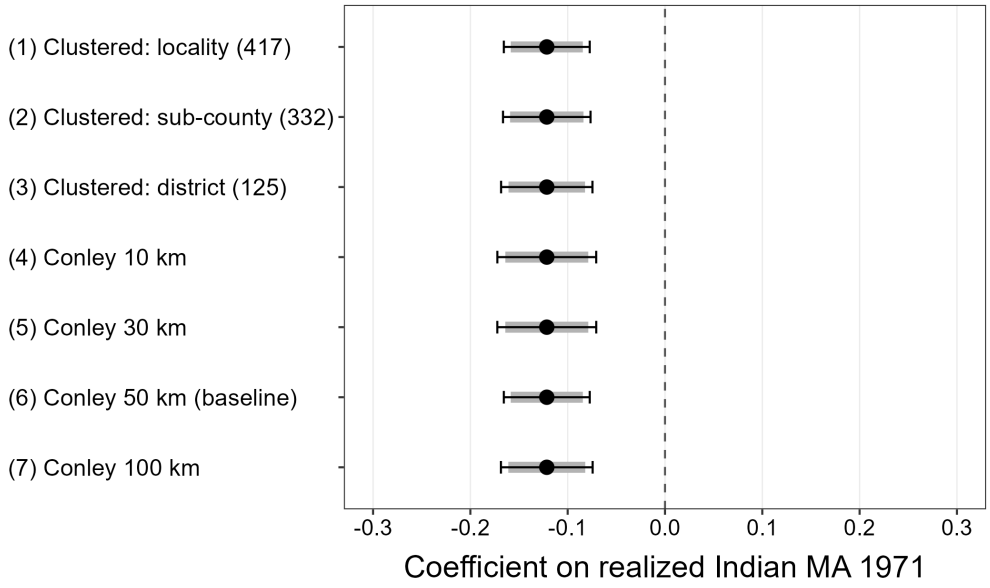
Figure A4: Alternative Exposure Measures and Decay Parameters



Notes: The figure reports the coefficient on realized Indian market access in 1971 multiplied by the standard deviation of realized market access under each row's measure, where the dependent variable is the log-transformed crop value of farmland and every regression is the specification of column 3 of Table 1. The first row is the baseline, a gravity sum over all 1959 Indian-populated towns with a decay parameter θ of 10 km. The second row replaces the gravity sum with its percentile rank; the third replaces it with exposure to the nearest town's Indian population; the remaining rows vary θ from 20 to 50 km, re-solving the gravity measure at the counterfactual road-completion schedules. Thick and thin bars are 90 and 95 percent confidence intervals, based on standard errors allowing for spatial correlation among observations within 50 kilometers following Conley (1999). All regressions exclude localities within 1959 towns.

Alternative inference. Appendix Figure A5 varies the variance estimator. Clustering at the locality (417 clusters), sub-county (332) or district (125) gives standard errors of 0.022 to 0.024. Conley standard errors with cutoffs of 10, 30 and 100 kilometers give 0.024 to 0.026, against the baseline 0.023 at 50 kilometers.

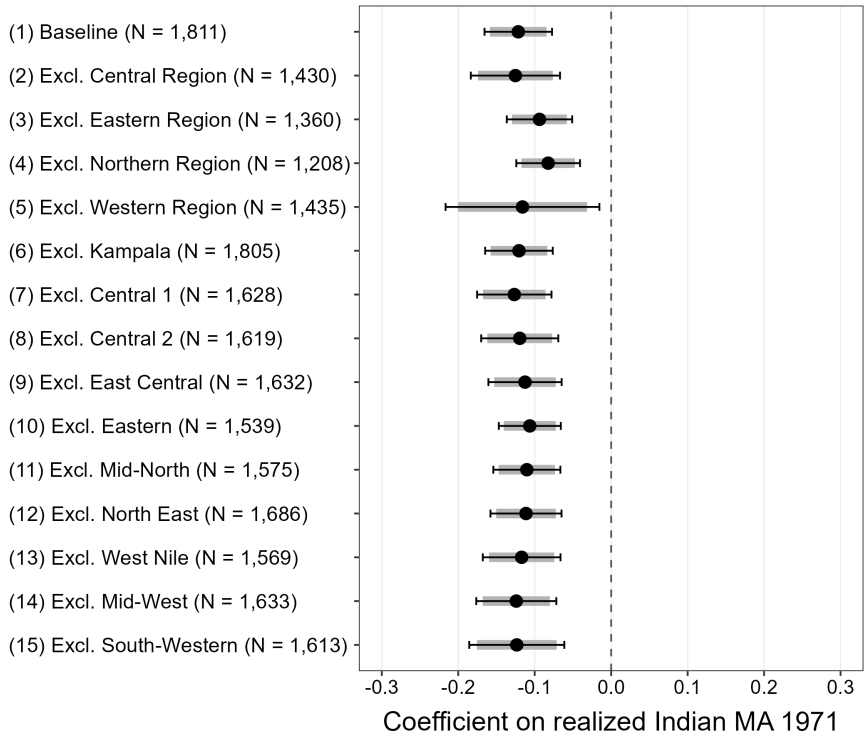
Figure A5: Alternative Inference



Notes: The figure reports the coefficient on realized Indian market access in 1971 from the specification of column 3 of Table 1, with standard errors clustered at the locality, sub-county, or district level (number of clusters in parentheses) or allowing for spatial correlation within 10, 30, 50, or 100 kilometers following Conley (1999). Thick and thin bars are 90 and 95 percent confidence intervals. All regressions exclude localities within 1959 towns.

Excluding one region or sub-region at a time. Appendix Figure A6 re-estimates the specification excluding one of Uganda's four regions or ten sub-regions at a time.

Figure A6: Excluding One Region or Sub-region At A Time



Notes: The figure plots the coefficient on realized Indian market access in 1971 from Equation 2, where the dependent variable is the log-transformed crop value of farmland. Every row is the specification of column 3 of Table 1, excluding the region or sub-region indicated. Thick and thin bars are 90 and 95 percent confidence intervals, based on standard errors allowing for spatial correlation among observations within 50 kilometers following Conley (1999). All regressions exclude localities within 1959 towns.

Indian versus general market access. Appendix Table A6 restricts the sample to households whose nearest 1959 town, under 1971 travel costs, had a low Indian share of its population. At cutoffs of 5 and 10 percent the coefficients are positive and insignificant. At cutoffs of 15 and 20 percent, they are negative, small, and insignificant. The results show no sign of an effect where road upgrads improved access to towns with few Indians.

Table A6: Indian Expulsion and Crop Value of Farmland by Indian Share of the Nearest Town

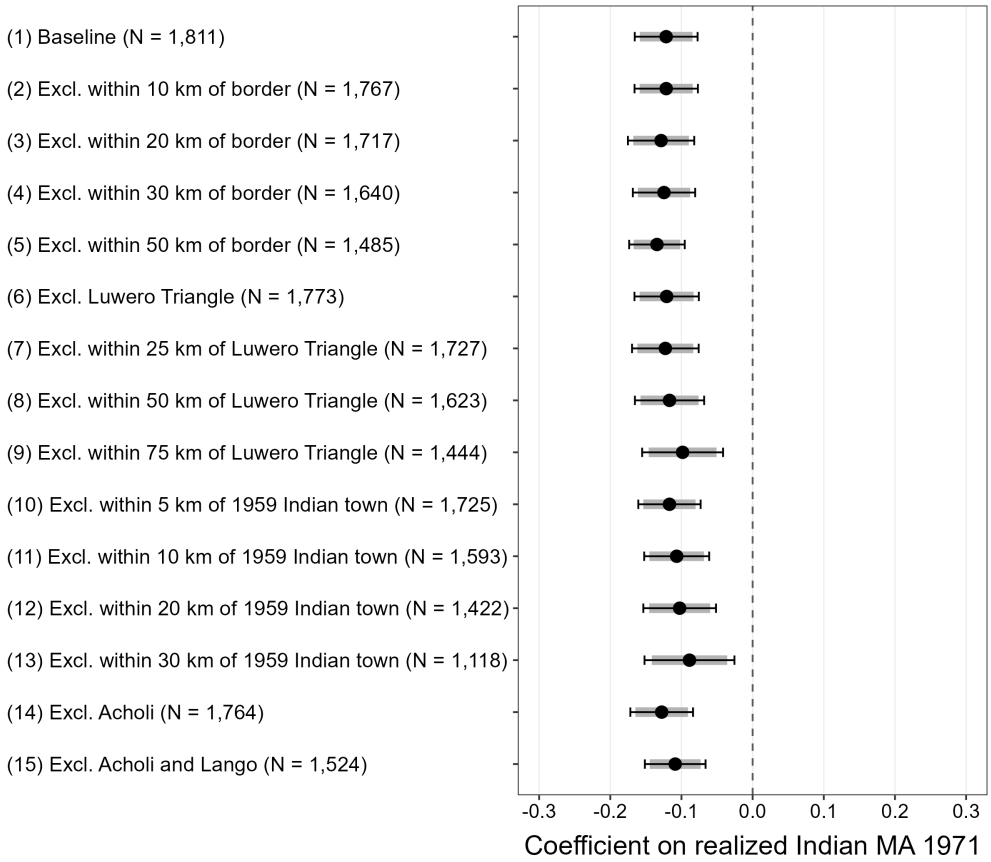
	≤5% (1)	≤10% (2)	≤15% (3)	≤20% (4)
Log crop value of farmland				
Realized Indian MA 1971	0.039	0.093	-0.034	-0.035
	(0.045)	(0.059)	(0.080)	(0.080)
Expected Indian MA 1971	-0.049	-0.062	0.022	0.023
	(0.065)	(0.060)	(0.082)	(0.082)
Observations	183	315	576	598
Localities	35	64	146	153
R^2	0.426	0.411	0.238	0.237
Mean of dependent variable	12.694	13.048	12.984	12.995
Agroclimatic and geographic controls	Y	Y	Y	Y
Household controls	Y	Y	Y	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is the log-transformed crop value of farmland, on households whose nearest 1959 town under 1971 travel costs has an Indian share of town population at or below the cutoff in the column heading. A town is classified only if its African population is recorded. All columns include agroclimatic, geographic and household controls and exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Excluding border, conflict-affected, and town-adjacent localities. Appendix Figure A7 excludes localities exposed to shocks concentrated in particular areas. Excluding localities within 10, 20, 30 and 50 kilometers of the Kenyan and Tanzanian borders gives estimates between -0.121 and -0.134. Excluding the Luweo Triangle districts and localities within 25, 50, and 75 kilometers gives estimates between -0.098 and -0.123. Excluding Acholi, the area most affected by the insurgency in the North, gives -0.128, and excluding both Acholi and Lango yields -0.109.

Figure A7 also excludes localities within 5, 10, 20 and 30 kilometer-equivalents of travel on the 1971 network from the nearest 1959 Indian-populated town. The estimates decline gradually, from -0.117 to -0.089. At the widest radius, which drops 38 percent of the sample, the decline remains 13.0 percent and significant at the 1 percent level.

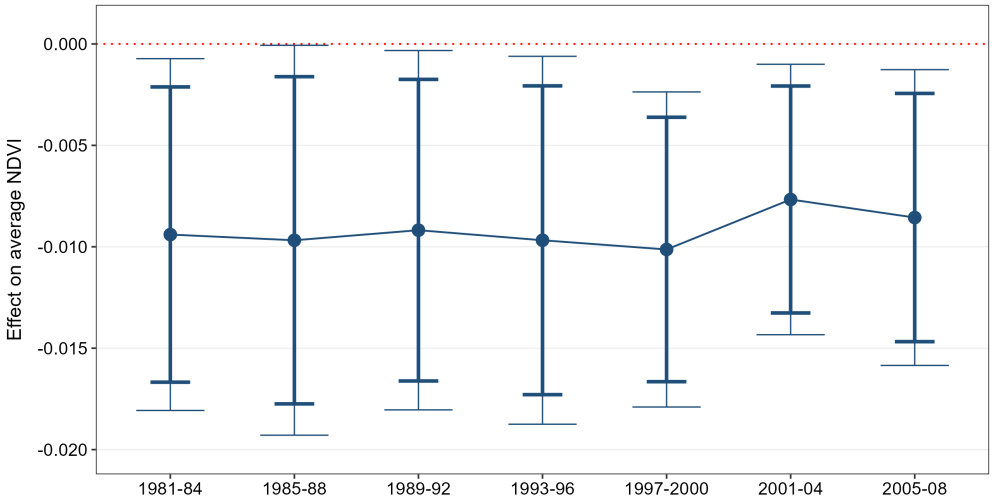
Figure A7: Excluding Border, Conflict-Affected and Town-Adjacent Localities



Notes: The figure reports the coefficient on realized Indian market access in 1971 from the specification of column 3 of Table 1. The first row is the baseline. Subsequent rows exclude localities within the indicated distance of the Kenyan or Tanzanian border; the districts of the Luwero Triangle (Luweero, Nakaseke and Nakasongola) and localities within the indicated distance of them; localities within the indicated least-cost travel distance, in kilometer-equivalents on the 1971 road network, of the nearest Indian-populated 1959 town; and the districts of Acholi, and of Acholi and Lango. Thick and thin bars are 90 and 95 percent confidence intervals, based on standard errors allowing for spatial correlation among observations within 50 kilometers following Conley (1999). All regressions exclude localities within 1959 towns.

Persistence over time. Appendix Figure A8 estimates Equation 2 for the 1964 gazetteer settlements, with average NDVI in each four-year period from 1981–1984 to 2005–2008 as the dependent variable. The specification includes expected exposure and the agroclimatic and geographic controls; the settlement-level data contain no household controls. The coefficient on realized exposure ranges from -0.008 to -0.010 NDVI units per log point and is significant at the 5 percent level in every period. The estimates show no trend over time, suggesting that the reduction in vegetation was already in place by the early 1980s and persisted through 2008.

Figure A8: Indian Expulsion and Agricultural Production, 1981–2008



Notes: The unit of analysis is a populated place recorded in the 1964 gazetteer. Each point is the coefficient on realized log Indian market access in 1971 from a separate cross-sectional regression, one per period. The dependent variable is average NDVI over the four years of the period. Every regression includes expected log Indian market access in 1971, the number of populated places within 10 km in 1964, ruggedness, rainfed cotton and coffee suitability, average annual rainfall over 1960–1969, and distances to the nearest 1959 town, to Kampala, and to the nearest mission. Thick bars are 90 percent and thin bars 95 percent confidence intervals, constructed from standard errors allowing for spatial correlation among observations within 50 km following Conley (1999). All regressions exclude settlements inside 1959 towns.

D Alternative Explanations

This appendix reports the evidence summarized in Section 5.3. The lower crop value of farmland in more-exposed localities could reflect four channels other than the loss of intermediation: differences in land use, a general contraction of the local economy, a loss of credit and inputs, and selective out-migration. I examine each in turn. Unless noted otherwise, all estimates use the specification of column 3 of Table 1.

Differences in land use. Table A7 decomposes the main outcome. By construction, log crop value of farmland equals the sum of three terms: log farmland in acres (Panel A), log crop value per cropped acre (Panel B), and log cropped area as a share of farmland (Panel C). I detailed the definition of each variable in Appendix A. Since the three regressions share the same sample and regressors, their coefficients add up exactly to the coefficient in Table 1.

Table A7: Decomposing the Effect of Indian Expulsion on Crop Value of Farmland

	(1)	(2)	(3)
Panel A. Log farmland (acres)			
Realized Indian MA 1971	-0.023	-0.024	-0.035
	(0.031)	(0.035)	(0.034)
Expected Indian MA 1971	0.006	0.016	0.028
	(0.031)	(0.037)	(0.036)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.012	0.068	0.179
Mean of dependent variable	0.916	0.916	0.916
Panel B. Log crop value per cropped acre			
Realized Indian MA 1971	-0.133***	-0.078***	-0.080***
	(0.020)	(0.027)	(0.026)
Expected Indian MA 1971	0.140***	0.058*	0.062*
	(0.020)	(0.034)	(0.034)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.049	0.110	0.120
Mean of dependent variable	12.933	12.933	12.933
Panel C. Log cropped area as a share of farmland			
Realized Indian MA 1971	-0.038**	-0.008	-0.007
	(0.016)	(0.022)	(0.022)
Expected Indian MA 1971	0.054***	0.015	0.014
	(0.017)	(0.024)	(0.024)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.072	0.151	0.156
Mean of dependent variable	-0.402	-0.402	-0.402
Agroclimatic and geographic controls	–	Y	Y
Household controls	–	–	Y

Notes: The table decomposes the estimates in Table 1: log crop value of farmland is the sum of the log of farmland (Panel A), log crop value per cropped acre (Panel B) and the log of cropped area as a share of farmland (Panel C), so the coefficients in each column sum to those in Table 1. Cropped area is the area of farmland planted, averaged over the two 2011 cropping seasons; rented-in parcels, for which the planted share is not recorded, are treated as fully planted. The explanatory variable is locality l's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. All regressions exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

General contraction of the local economy. Table A8 tests whether exposure reduced non-agricultural activity, using nighttime lights at the 1964 gazetteer settlements. The data are the stable-lights composites of the Defense Meteorological Satellite Program's Operational Linescan System (DMSP-OLS) for 1992, 2000, and 2010. For each settlement, I average the digital number over a 5-kilometer buffer around the settlement and define an indicator, scaled by 100, for whether this mean is positive. I use the extensive margin because most settlements are unlit. Table A8 reports the effect of exposure on whether historical settlements have positive nighttime light within 5 kilometers. The coefficients on realized exposure are positive in every year and insignificant. Table A9 reports the effect on the share of lit cells within 5 kilometers of a historical settlement. Similarly, the coefficients are not statistically significant.

Table A8: Indian Expulsion and Indicators for Lit Settlements

	(1)	(2)
Panel A. Lit in 1992 (×100)		
Realized Indian MA 1971	1.102	1.216
	(0.834)	(0.913)
Expected Indian MA 1971	-0.271	-0.624
	(0.574)	(0.731)
Observations	3,429	3,429
R^2	0.053	0.120
Mean of dependent variable	4.899	4.899
Panel B. Lit in 2000 (×100)		
Realized Indian MA 1971	0.965	1.087
	(1.372)	(1.529)
Expected Indian MA 1971	0.810	0.045
	(1.138)	(1.288)
Observations	3,429	3,429
R^2	0.101	0.191
Mean of dependent variable	13.386	13.386
Panel C. Lit in 2010 (×100)		
Realized Indian MA 1971	1.570	1.817
	(1.572)	(1.747)
Expected Indian MA 1971	0.337	-0.560
	(1.273)	(1.497)
Observations	3,429	3,429
R^2	0.103	0.208
Mean of dependent variable	15.165	15.165
Agroclimatic and geographic controls	–	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is an indicator for a positive mean DMSP-OLS digital number within 5 km of the settlement, scaled by 100. The explanatory variable is settlement s's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. The unit of analysis is a populated place recorded in the 1964 gazetteer. All regressions exclude settlements inside 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A9: Indian Expulsion and the Share of Lit Area within 5 km

	(1)	(2)
Panel A. Share of lit cells within 5 km in 1992 (%)		
Realized Indian MA 1971	0.294	0.300
	(0.283)	(0.279)
Expected Indian MA 1971	-0.058	-0.159
	(0.165)	(0.212)
Observations	3,429	3,429
R^2	0.031	0.089
Mean of dependent variable	1.208	1.208
Panel B. Share of lit cells within 5 km in 2000 (%)		
Realized Indian MA 1971	0.648	0.683
	(0.651)	(0.683)
Expected Indian MA 1971	0.007	-0.305
	(0.425)	(0.538)
Observations	3,429	3,429
R^2	0.070	0.164
Mean of dependent variable	3.924	3.924
Panel C. Share of lit cells within 5 km in 2010 (%)		
Realized Indian MA 1971	0.878	0.931
	(0.790)	(0.851)
Expected Indian MA 1971	-0.107	-0.459
	(0.512)	(0.665)
Observations	3,429	3,429
R^2	0.082	0.191
Mean of dependent variable	4.636	4.636
Agroclimatic and geographic controls	–	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is the percentage of DMSP-OLS 30-arc-second cells within 5 km of the settlement with a positive digital number. The explanatory variable is settlement s's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. The unit of analysis is a populated place recorded in the 1964 gazetteer. All regressions exclude settlements inside 1959 towns and those not connected to the road network. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Credit and inputs. Table A10 reports effects on farm implements and Table A11 on purchased inputs. The data come from the UNPS farm implements module (AGSEC10), which lists 23 implement types and asks, for each, how many the household owns, their total estimated value, and whether the household used the item in the past 12 months. In Panel A, implement value is the sum of the reported value of all implements the household owns, divided by its farmland in acres. In Panel B, labor-saving instruments are ploughs, animal-drawn ploughs, wheelbarrows and sprayers. The indicator equals 100 if the household used any of them, so that borrowed or rented implements also count. The hoe is excluded from this indicator since 99.7 percent of households use one in the data. In Panel C, the number of implements is the number of hoes, ploughs, animal-drawn ploughs, wheelbarrows and sprayers the household owns, counting only items it reports using.

Table A11 reports effects on purchased chemical inputs. These come from the plot-level input modules for the first and second seasons of 2011 (AGSEC3A and AGSEC3B). For each plot, the survey records whether the household applied inorganic fertilizer and whether it applied a pesticide, and if so of which type. I define three indicators, each scaled by 100, for whether the household used inorganic fertilizer, an insecticide, or a growth regulator on any plot in either season.

Table A10: Indian Expulsion and Capital Investment and Modernization

	(1)	(2)	(3)
Panel A. Log implement value per acre of farmland			
Realized Indian MA 1971	-0.009	0.006	0.001
	(0.034)	(0.046)	(0.046)
Expected Indian MA 1971	0.031	-0.006	-0.001
	(0.039)	(0.055)	(0.055)
Observations	1,800	1,800	1,800
Localities	415	415	415
R^2	0.020	0.054	0.066
Mean of dependent variable	9.208	9.208	9.208
Panel B. Uses any labour-saving implement (×100)			
Realized Indian MA 1971	-0.231	-0.556	-0.914
	(1.002)	(0.792)	(0.691)
Expected Indian MA 1971	0.650	0.465	0.868
	(0.901)	(0.745)	(0.681)
Observations	1,805	1,805	1,805
Localities	416	416	416
R^2	0.005	0.067	0.119
Mean of dependent variable	27.535	27.535	27.535
Panel C. Log number of implements used			
Realized Indian MA 1971	-0.005	0.013	0.003
	(0.009)	(0.010)	(0.009)
Expected Indian MA 1971	0.013	-0.018*	-0.007
	(0.008)	(0.011)	(0.009)
Observations	1,790	1,790	1,790
Localities	414	414	414
R^2	0.008	0.052	0.293
Mean of dependent variable	1.163	1.163	1.163
Agroclimatic and geographic controls	–	Y	Y
Household controls	–	–	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is the log-transformed value of farm implements per acre of farmland in Panel A, an indicator for using any labour-saving implement in Panel B, and the log-transformed number of implements the household uses in Panel C. Implement value is the reported value of all farm implements the household owns. Labour-saving implements are ploughs, animal-drawn ploughs, wheel barrows and sprayers; the indicator is scaled by 100. The explanatory variable is locality l's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. All regressions exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table A11: Indian Expulsion and the Use of Purchased Agricultural Inputs

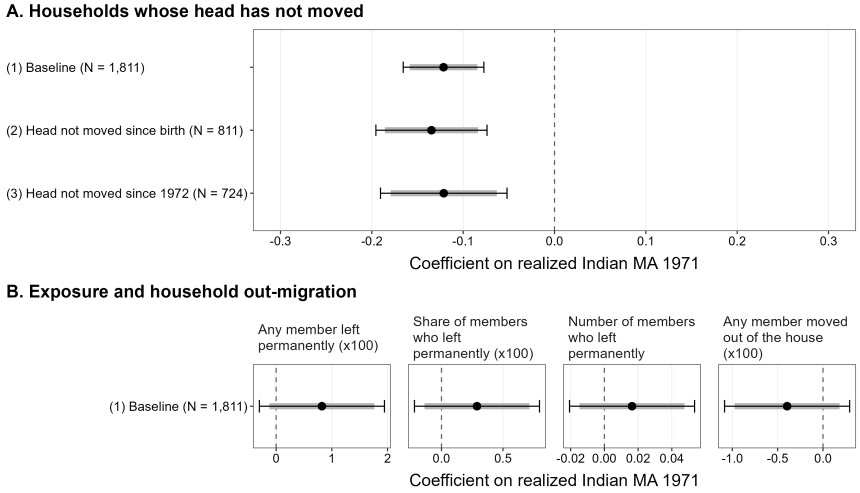
	(1)	(2)	(3)
Panel A. Uses inorganic fertiliser (×100)			
Realized Indian MA 1971	-0.229	0.174	0.114
	(0.677)	(0.761)	(0.754)
Expected Indian MA 1971	0.283	-0.067	0.014
	(0.612)	(0.830)	(0.819)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.001	0.039	0.046
Mean of dependent variable	5.743	5.743	5.743
Panel B. Uses insecticide (×100)			
Realized Indian MA 1971	-0.108	-0.433	-0.498*
	(0.325)	(0.303)	(0.282)
Expected Indian MA 1971	0.492	0.722*	0.816**
	(0.306)	(0.387)	(0.366)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.010	0.029	0.035
Mean of dependent variable	8.614	8.614	8.614
Panel C. Uses a growth regulator (×100)			
Realized Indian MA 1971	-0.850	0.083	0.037
	(0.915)	(0.654)	(0.654)
Expected Indian MA 1971	0.826	-0.084	-0.022
	(0.824)	(0.690)	(0.687)
Observations	1,811	1,811	1,811
Localities	417	417	417
R^2	0.003	0.104	0.112
Mean of dependent variable	5.356	5.356	5.356
Agroclimatic and geographic controls	–	Y	Y
Household controls	–	–	Y

Notes: The table reports estimates of Equation 2, where the dependent variable is an indicator for using inorganic fertiliser in Panel A, an insecticide in Panel B and a growth regulator in Panel C, on any plot in either season; all are scaled by 100. Households without a plot-input record are excluded rather than coded as non-users. The explanatory variable is locality l's log Indian market access in 1971, reported alongside its expected value across counterfactual road-completion schedules. All regressions exclude localities within 1959 towns. Standard errors allowing for spatial correlation among observations within 50 km, following Conley (1999), are reported in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Selective out-migration. Figure A9, Panel A reports the main estimate in samples restricted by household head's residence history. The household questionnaire asks each member how many years they have lived in the current village or community, with a separate code for having always lived there. A head has lived in the community since birth if they report always having lived there. This represents 811 households in the data. A head has lived in the community since 1972 if they report at least 39 years of residence, or always having lived there and are at least 39 years of residence, or always having lived there and are at least 39 years old. This represents 724 households.

Panel B uses the baseline sample with out-migration from the household as the outcome. The household roster records former members together with the reason they are no longer present. I count a member as having left permanently if the roster records them as having left permanently. Fromn this, I construct three outcomes: an indicator for any member having left permanently, the share of roster members who left permanently, and their number. A fourth outcome is an indicator for any member having moved out of the dwelling.

Figure A9: Selective Out-Migration

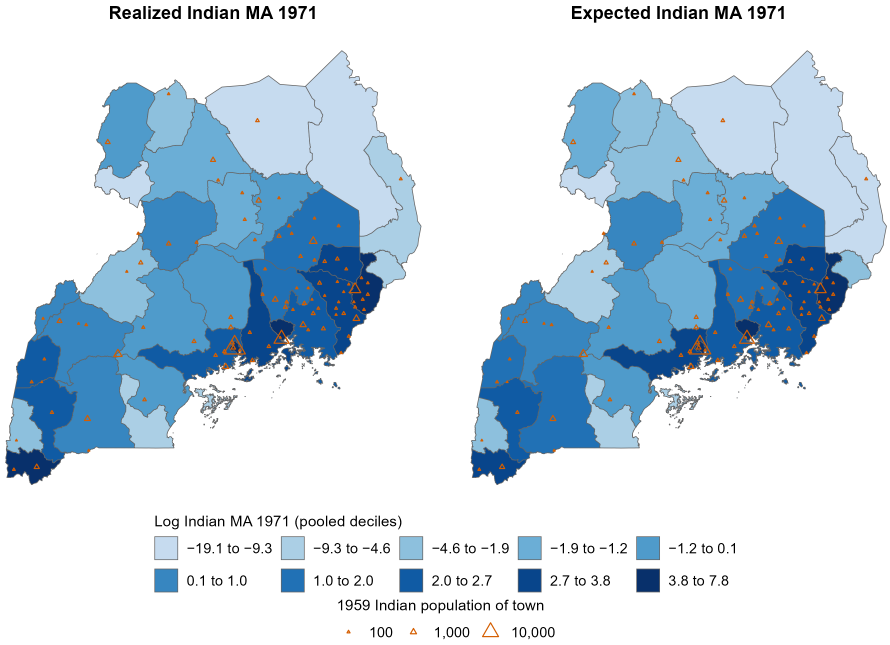


Notes: The figure reports the coefficient on realized Indian market access in 1971 from the specification of column 3 of Table 1. Panel A shows the impact on log crop value of farmland for the baseline sample and for households whose head has lived in the community since birth, or since 1972; six households with missing residence history are excluded from both restricted samples. Panel B uses the baseline sample with out-migration as the dependent variable: an indicator for any household member having left permanently, the share of the roster who left permanently, their number, and an indicator for any member having moved out of the dwelling; indicators and shares are scaled by 100. Residence and out-migration are recorded in 2011 and are therefore post-treatment. Thick and thin bars are 90 and 95 percent confidence intervals, based on standard errors allowing for spatial correlation among observations within 50 kilometers following Conley (1999). All regressions exclude localities within 1959 towns.

Loss of human capital. This section examines whether the expulsion lowered the schooling of children who were of school age in 1972, a channel consistent with the lower education spending reported. I use the 1991 Population and Housing Census, distributed by IPUMS-International. All estimates use cells of district of birth by year of birth, weighted by the number of persons reporting schooling, with standard errors clustered by district of birth.

The census records each person's age, district of birth, and highest grade completed. I keep persons born in Uganda and assign year of birth as 1991 minus age. Years of schooling is the highest grade completed. A person has completed primary school if they report at least seven years of schooling. I measure exposure at the district of birth. Realized and expected Indian market access in 1971 for a district are the unweighted means of the settlement-level measures over the 1964 settlements in that district. Figure A10 illustrates the variation across districts of birth.

Figure A10: Realized and Expected Indian Market Access Across Districts of Birth



Notes: The figure shows log Indian market access in 1971 for the 34 districts of birth recorded in the 1991 census. Expected market access is the mean over 1,000 counterfactual road-completion schedules. Triangles mark 1959 towns with an Indian population, sized by Indian population.

Table A12 examines whether children who were of school age at the expulsion completed less schooling. The regression compares treated and control cohorts across districts of birth with different realized Indian market access in 1971, conditional on expected Indian market access, district-of-birth fixed effects and year-of-birth fixed effects. In column (1), one log point higher realized Indian market access lowers schooling

among treated cohorts by 0.12 years, or about 3.8 percent of the control-cohort mean. In column (3), it lowers the share completing primary school by 1.5 percentage points, or 7.5 percent of the 20 percent control mean. In columns (2) and (4), I additionally include region by year-of-birth fixed effects, which absorb differential schooling expansion across Uganda's four regions. The coefficients do not change much. The randomization-inference intervals exclude zero in every column.

Table A12: Indian Market Access and Schooling by Birth Cohort

	Mean years of schooling		Completed primary (%)	
	(1)	(2)	(3)	(4)
Realized Indian MA × Treated	-0.120**	-0.114*	-1.511**	-1.435*
	(0.058)	(0.062)	(0.690)	(0.768)
	[-0.217, -0.028]	[-0.217, -0.005]	[-2.624, -0.513]	[-2.696, -0.243]
Expected Indian MA × Treated	0.147**	0.143**	1.785***	1.736**
	(0.057)	(0.061)	(0.644)	(0.759)
Birth district FE	Y	Y	Y	Y
Birth year FE	Y	Y	Y	Y
Region × birth year FE	–	Y	–	Y
Mean, control cohorts	3.14	3.14	20.04	20.04
Observations (individuals)	519,640	519,640	519,640	519,640
Birth district × year cells	1,360	1,360	1,360	1,360
Birth districts	34	34	34	34

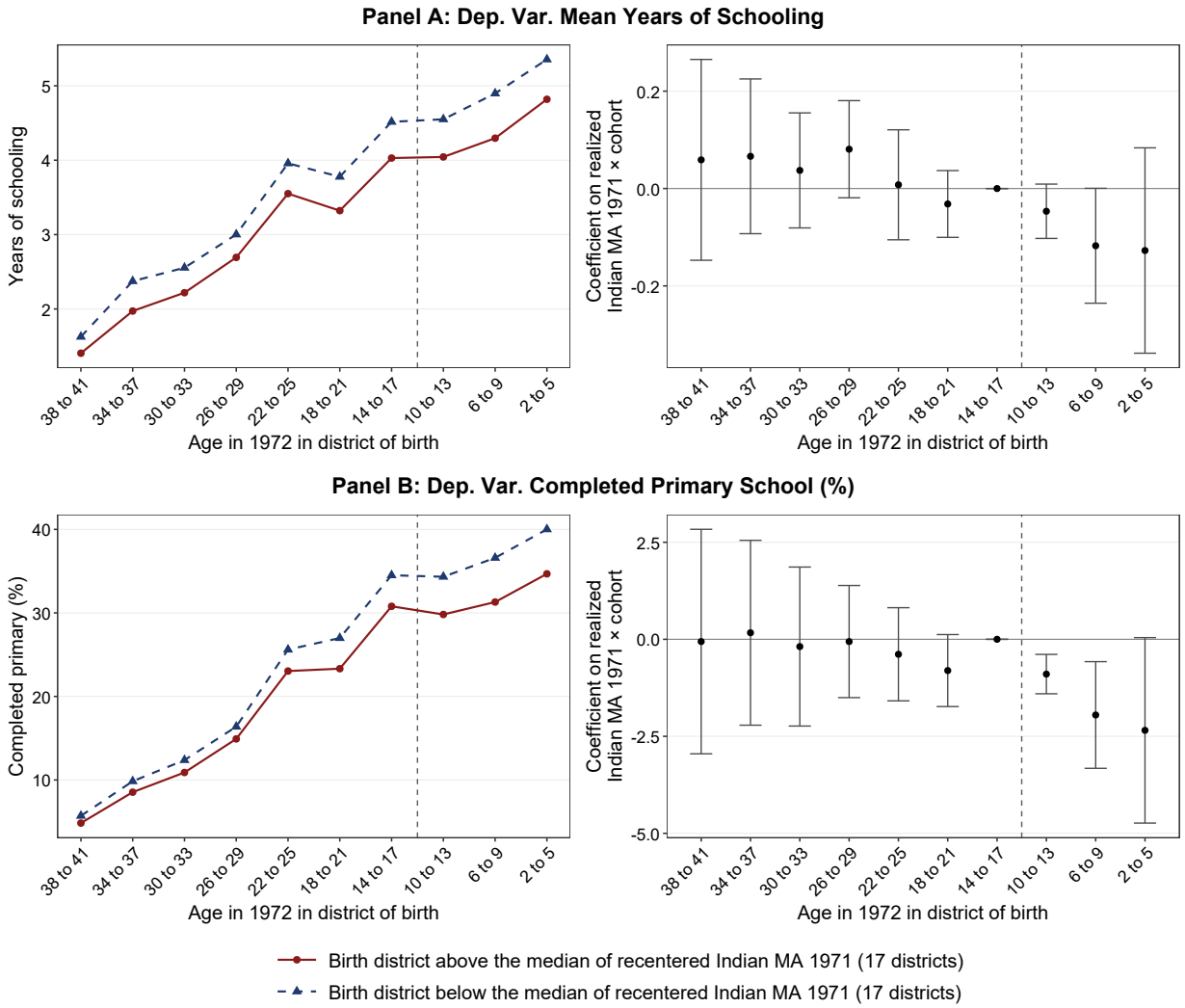
Notes: The table reports cohort difference-in-differences estimates on district-of-birth × year-of-birth cells from the 1991 census, for Ugandan-born persons born 1931–1970. The dependent variable is mean years of schooling in columns 1–2 and the percentage who completed primary school in columns 3–4. The explanatory variable is log Indian market access in 1971 in the district of birth, interacted with an indicator for cohorts aged 13 or younger in 1972, controlling for its expected value interacted with the same indicator. Control cohorts were aged 14–41 in 1972. Cells are weighted by the number of persons reporting schooling. Regions are Uganda's four regions. Standard errors clustered by district of birth are reported in parentheses. Randomization-inference 95% confidence intervals following Borusyak and Hull (2023) are reported in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Indian Market Access and Sector of Employment by Birth Cohort

	Share of employment (%)		
	Agriculture (1)	Manufacturing (2)	Services (3)
Panel A: Birth district and birth year FE			
Realized Indian MA × Treated	1.173	-0.256	-0.917
	(1.019)	(0.214)	(0.876)
	[-0.020, 2.707]	[-0.656, 0.128]	[-2.131, 0.054]
Expected Indian MA × Treated	-1.214	0.253	0.962
	(0.931)	(0.191)	(0.818)
Panel B: Adding region × birth year FE			
Realized Indian MA × Treated	0.688	-0.353**	-0.335
	(0.767)	(0.146)	(0.650)
	[-0.396, 1.985]	[-0.619, -0.113]	[-1.382, 0.543]
Expected Indian MA × Treated	-0.654	0.392***	0.262
	(0.731)	(0.142)	(0.629)
Mean, control cohorts	77.75	5.24	17.01
Observations (individuals)	399,205	399,205	399,205
Birth district × year cells	1,360	1,360	1,360
Birth districts	34	34	34

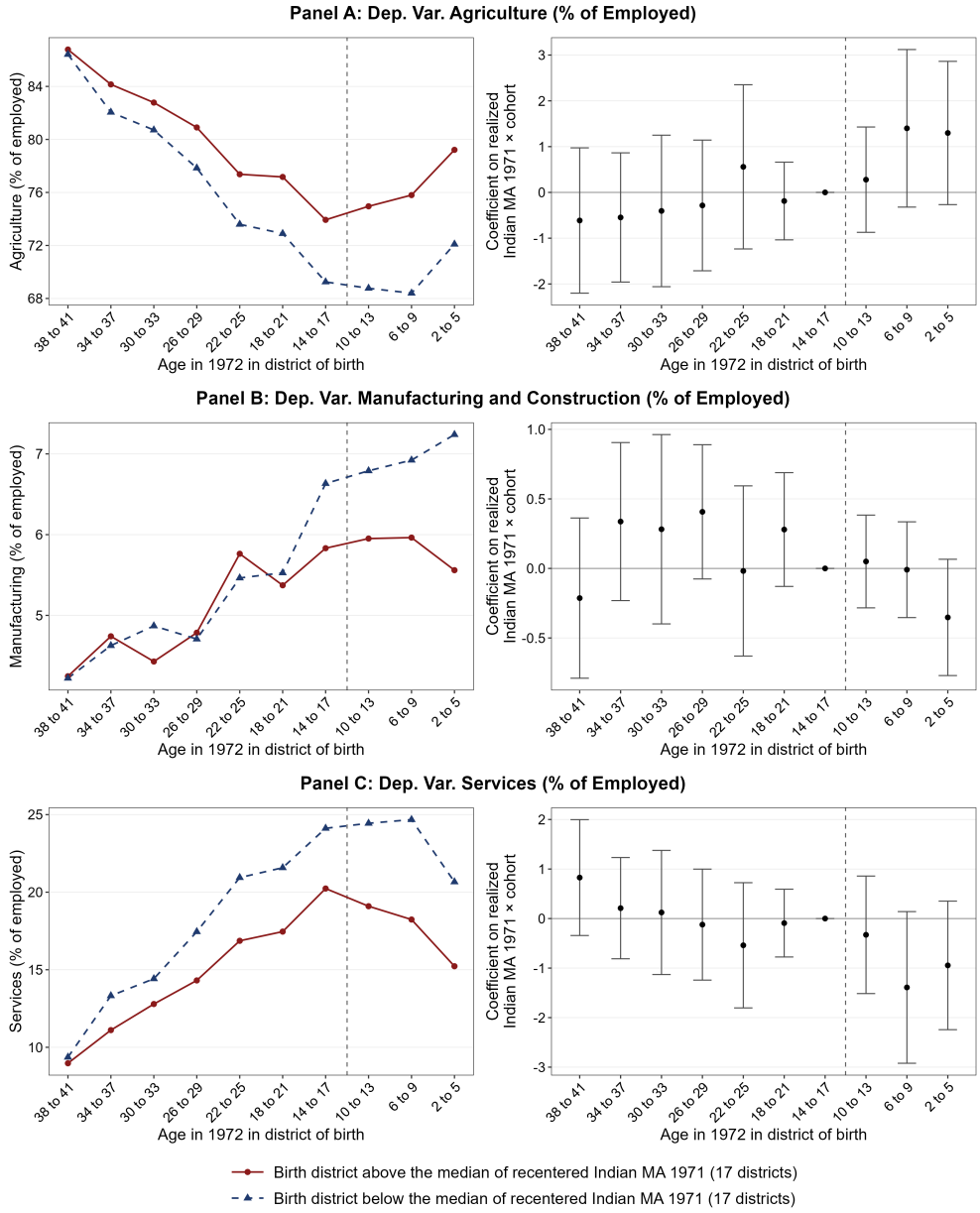
Notes: The table reports cohort difference-in-differences estimates on district-of-birth × year-of-birth cells from the 1991 census, for employed Ugandan-born persons born 1931–1970. The dependent variable is the percentage of the employed working in agriculture (column 1), manufacturing and construction (column 2) and services (column 3), classified by occupation. The explanatory variable is log Indian market access in 1971 in the district of birth, interacted with an indicator for cohorts aged 13 or younger in 1972, controlling for its expected value interacted with the same indicator. Control cohorts were aged 14–41 in 1972. Cells are weighted by the number of employed persons. Regions are Uganda's four regions. Standard errors clustered by district of birth are reported in parentheses. Randomization-inference 95% confidence intervals following Borusyak and Hull (2023) are reported in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A11: Indian Market Access and Schooling by Birth Cohort



Notes: The unit of analysis is district of birth × four-year birth cohort in the 1991 census, Ugandan-born. The left figure reports person-weighted means for birth districts above and below the median of recentered Indian market access 1971. The right figures report coefficients on realized Indian market access 1971 interacted with cohort dummies, controlling for expected market access interacted with the same dummies, district-of-birth and birth-year fixed effects. The reference cohort is aged 14–17 in 1972, the typical terminal age of formal schooling. The dashed line separates cohorts aged 14 and older in 1972 from those aged 13 and younger. 95% confidence intervals clustered by district of birth.

Figure A12: Indian Market Access and Structural Transformation by Birth Cohort



Notes: The unit of analysis is district of birth × four-year birth cohort in the 1991 census, Ugandan-born. The left figure reports person-weighted means for birth districts above and below the median of recentered Indian market access 1971. The right figures report coefficients on realized Indian market access 1971 interacted with cohort dummies, controlling for expected market access interacted with the same dummies, district-of-birth and birth-year fixed effects. The reference cohort is aged 14–17 in 1972, the typical terminal age of formal schooling. The dashed line separates cohorts aged 14 and older in 1972 from those aged 13 and younger. 95% confidence intervals clustered by district of birth.